Journal of King Saud University Computer and Information Sciences (2025) 38:21
https://doi.org/10.1007/s44443-025-00334-6

REVIEW PAPER

Improving negative rejection ability in language models: A review of fine-tuned LLMs, RAG, and RAFT

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Received: 3 July 2025 / Accepted: 10 October 2025 / Published online: 6 December 2025

Abstract
Large Language Models (LLMs) excel in text understanding and generation but struggle to reject irrelevant, ambiguous, or misleading queries, termed negative rejection, impacting reliability in high-stakes contexts. This paper reviews negative rejection, analyzing three approaches: fine-tuned LLMs, Retrieval-Augmented Generation (RAG), and Retrieval-Augmented Fine-Tuning (RAFT). Through a literature review, we assess their rejection accuracy, hallucination rates, computational costs, and domain adaptability. We introduce a four-dimensional framework—architecture, metrics, domain application, and dataset diversity—to organize existing studies and benchmarks for comparative analysis (Cao, 2024). Our findings show fine-tuning boosts specificity but lacks universality, RAG enhances context but suffers from retrieval noise, and RAFT balances rejection and efficiency. We recommend future LLMs embed rejection awareness in retrieval and generation. This work synthesizes current research, offering analytical insights into negative rejection strategies and guiding NLP system development.

Keywords Negative rejection · LLM safety · Fine-tuning · RAG · RAFT · Hallucination mitigation

1 Introduction

1.1 Aim and scope of the survey

Large Language Models (LLMs) such as BERT and GPT have revolutionized natural language processing (NLP) in language translation, question-answering, and conversational AI. For example, BERT's bidirectional contextual representation has been remarkable in Named Entity Recognition (NER), sentiment analysis, and text classification with state-of-the-art accuracy on widely used benchmarks (Salıcı & Ölçer 2024). Similarly, the autoregressive structure of GPT has delivered impressive capabilities for text generation and is regarded amongst the best from a creative writing or chatbot perspective, and iterations thereof as a general open language modeling task (Salıcı & Ölçer 2024). To a considerable extent, these capabilities can be traced back to the underlying Transformer architecture that leverages self-attention to process contextual dependencies and combines the cognitive development of language in the depth of data processing at scale (Salıcı & Ölçer 2024). That said, LLMs struggle with irrelevant, ambiguous, or adversarial inputs and the inability to reject answering inputs if there's no sufficient evidence and information. This major limitation is particularly important in situations that undermine reliability and trustworthiness if the LLMs are unaware of frameworks, and for any occasion that carries substantial significance. The ability to reject answering questions when there is no available and relevant information and context is called negative rejection ability (Chen et al. 2024).

This survey aims to examine how current LLMs deal with these challenging situations. Looking for discussion on existing approaches to enhancing negative rejection abilities and where these approaches leave off. We involve discussion of architecture, training, and assessment metrics regarding negative rejection abilities. By reviewing current developments and recommending topics for future research on enhancing LLMs toward more reliable and trustworthy behavior, we hope to contribute to the development of more reliable and trustworthy LLMs. This survey contributes to the growing body of research advocating for more responsible AI behaviors. Previous studies have highlighted that

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negative rejection is not merely a technical safeguard, but a foundational requirement for trustworthy and privacy-preserving intelligent systems, particularly in domains where false responses may lead to significant harm (Ali et al. 2025).

1.2 Background and motivation

1.2.1 Importance of negative rejection in LLMs

Machines need to reject queries that do not fit the task well in order to prevent them from giving inaccurate answers (Chen et al. 2024). Sometimes, ineffective rejection brings about hallucinations that convince people, mainly in high-stress areas like healthcare, law and finance (García-Ferrero et al. 2023; Feldman et al. 2024). For instance, a weak rejection of information may misinterpret case laws in law or give incorrect medical recommendations in medicine (Ziaei & Schmidgall 2023).

This follows the main principle of AI ethics which encourages systems to avoid acting in sensitive situations to protect trust and boundaries (Ali et al. 2025). The difficulties involved in negation and meeting compliance show the need for well-planned rejection mechanisms (Truong et al. 2023).

1.2.2 Role of fine-tuning and retrieval-augmented methods

Tuning an LLM by using labeled data helps it perform a specific task, but as the size of the model grows, the process becomes too expensive, as shown in Fig. 1. To improve, Parameter-Efficient Fine-Tuning (PEFT) methods like adapters, LoRA and prompt-tuning update just a few parameters to balance their effectiveness and usage of resources (Han et al. 2024). Nevertheless, these techniques regularly miss out on filtering unnecessary context which can cause hallucinations and uncertain results (Chen et al. 2023; Gekhman et al. 2024).

Novel approaches combine cross-entropy loss with auxiliary reconstruction loss, improving distinction between relevant and irrelevant contexts. For instance, constraining in-scope embeddings in latent space enhances out-of-scope (OOS) detection by 3.22% (AUPR) on benchmarks like StackOverflow without compromising intent classification (Zhang et al. 2024a). While PEFT reduces costs, it does not have systems to guarantee that responses are valid in emergency situations (Han et al. 2024).

Figure 2 shows that Retrieval-Augmented Generation (RAG) integrates external documents to ground responses in factual context (Gao et al. 2023). However, it risks propagating irrelevant or noisy retrieved information, highlighting the need for robust retrieval filtering and hybrid approaches (Table 1).

These methods alone insufficiently reject irrelevant or ambiguous context, often leaving models prone to hallucinations (Chen et al. 2023; Gekhman et al. 2024). Novel approaches combine cross-entropy loss with reconstruction loss to better distinguish relevant from irrelevant contexts. For example, constraining in-scope embeddings in latent space improves separation from out-of-scope (OOS) samples, enhancing OOS detection by 3.22% (AUPR) on the StackOverflow dataset without compromising intent classification (Zhang et al. 2024a). While PEFT reduces computation, it lacks mechanisms to prevent hallucinations or ensure response validity, critical for safety-sensitive applications (Han et al. 2024).

Retrieval-Augmented Generation (RAG) integrates external documents to ground responses in factual context (Gao et al. 2023).

RAG may address factual grounding and contextual relevance, but it brings new challenges of propagating irrelevant information retrieved in the process or being able to delineate knowledge that may be helpful from noise. Thus, RAG improves the model's ability to respond to knowledge-intensive tasks but is limited in improving the model's negative rejection ability. The recent work, such as Retrieval-Augmented Dual Instruction Tuning (RA-DIT), seeks to improve retrieval in LLMs (Lin et al. 2023).

RAFT is an extension of RAG, adding fine-tuning models on "golden" and distractor documents, to more directly specialize models to reject unhelpful contexts (Zhang et al. 2024b). RAFT trains models explicitly to reject unhelpful inputs, increasing the model's ability to distinguish relevant and irrelevant context. However, RAFT faces several practical limitations in real-world deployments:

Computational Expense: The integration of retrieval mechanisms with fine-tuning significantly increases the computational overhead compared to standard RAG or fine-tuning techniques. RAFT requires more resources during training, as it incorporates large external knowledge retrieval into the fine-tuning process. This makes RAFT computationally expensive and less scalable, especially in resource-constrained environments (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e).

Latency Issues: Real-time applications may experience increased latency due to the additional retrieval step. While the fine-tuning phase benefits from retrieval awareness, inference time could be impacted by retrieval delays, especially in environments where rapid responses are critical, such as conversational AI or real-time medical decision-making (Fang et al. 2024a, b).

Domain-Specific Overfitting: Although RAFT improves rejection ability by filtering out irrelevant contexts, it may struggle with domain-specific overfitting. The model may perform well on specialized tasks within well-defined

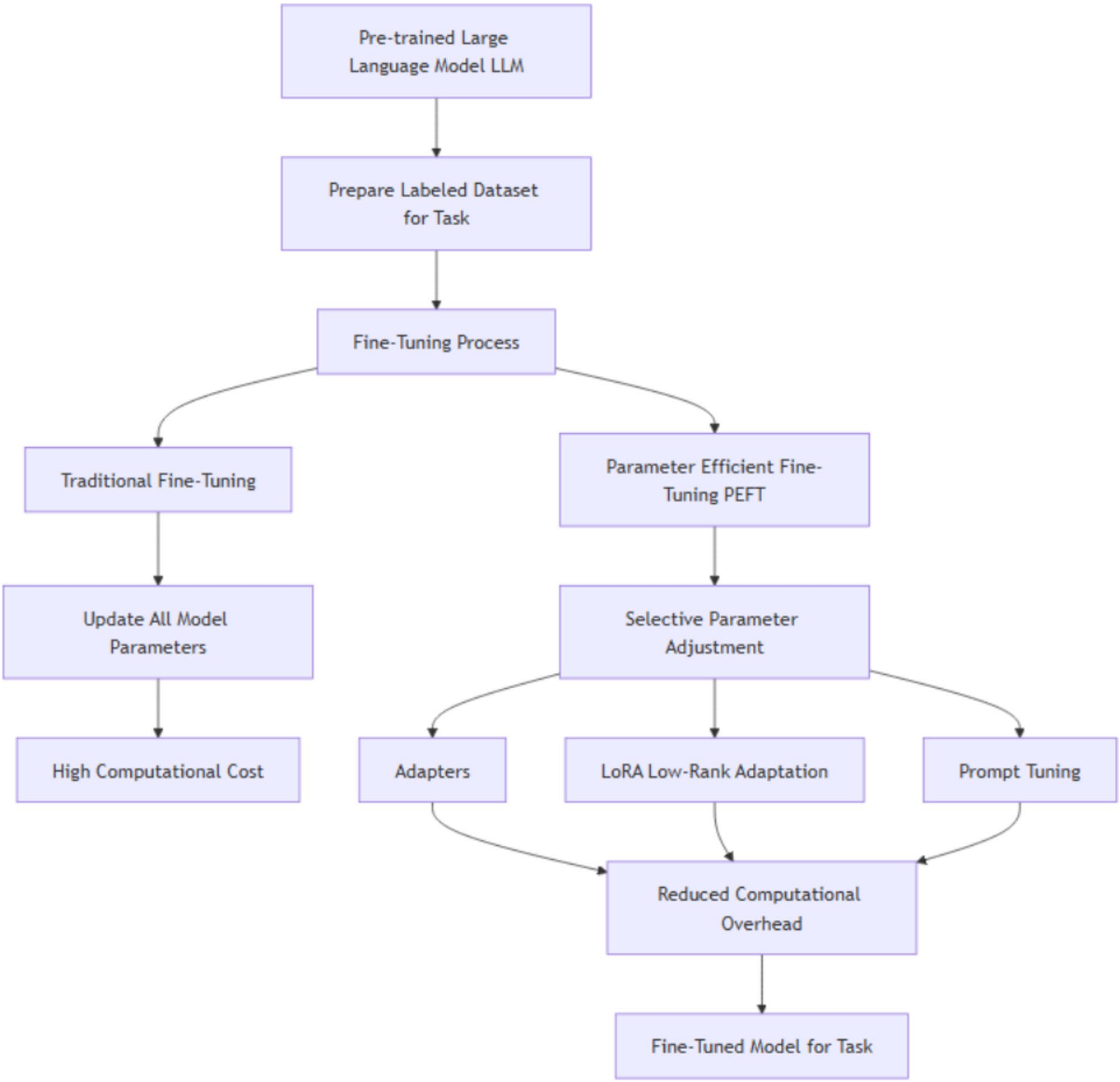

Fig. 1 Fine-tuning process

domains but may not generalize effectively to other domains with less relevant training data (Oliveira et al. 2024).

As Fig. 3 shown, RAFT preserves the advantages of RAG while also attempting to mitigate some of its disadvantages by training the model to respond meaningfully in cases where retrieval is uncertain or messy. RAFT also employed contrastive learning and chain-of-thought prompting to improve the quality of answers and the degree of rejection. Results substantiate that RAFT improved negation and rejection performance with LLM (Zhang et al. 2024b).

1.3 Research objectives

This research investigates enhancing LLMs' ability to reject irrelevant, ambiguous, or misleading inputs, termed negative rejection, a critical limitation in high-risk applications (García-Ferrero et al. 2023). We analyze three methods—fine-tuning, RAG, and RAFT—to improve negative rejection.

The study aims to: (1) define negative rejection as a measurable performance dimension; (2) compare these methods' rejection accuracy, hallucination mitigation,

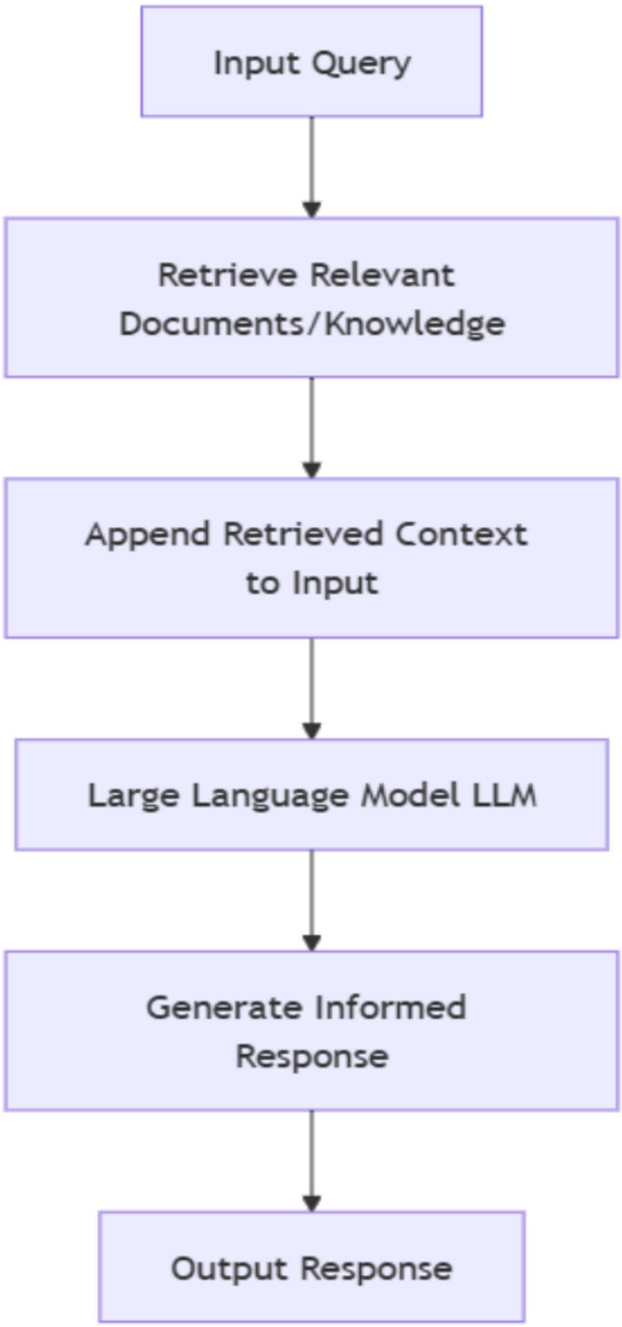

Fig. 2 RAG implementation

computational cost, and domain adaptability; (3) develop a four-dimensional framework—model architecture, evaluation metrics, application domains, and dataset diversity—to synthesize existing literature and benchmarks for analysis (Cao 2024). This review provides conceptual clarity and practical guidance for safer LLMs, highlighting trade-offs and limitations to guide future rejection mechanism development.

1.4 Organization of the paper

The structure of this paper is arranged as follows. Section 2 introduces the Survey Methodology, mainly elaborating the systematic process of literature selection, inclusion criteria and data extraction. Section 3 is a literature review, which outlines the key concept of negative rejection ability and analyzes the three cutting-edge technologies of Fine-Tuning, RAG and RAFT. Section 4 conducts a comparative analysis to summarize the advantages, limitations and applicable scenarios of each of the above-mentioned technologies. Section 5 explores challenges and open questions, covering technical, theoretical and ethical considerations in the development of negative rejection mechanisms. Section 6 proposes future research directions, including suggestions such as improving retrieval efficiency, cross-domain adaptability, and integrating structured knowledge. Finally, the paper makes a conclusion and presents the research implications.

2 Survey methodology

This review follows the methodology of a Systematic Literature Review (SLR) to analyze the negative rejection ability of LLMs (Kitchenham 2004). Inspired by prior high-quality surveys (Chang et al. 2024; Min et al. 2023), we designed a structured search strategy to ensure comprehensive coverage. Searches combined keyword queries and citation tracking across major databases (IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar). The overall workflow—identification, screening, eligibility, and inclusion—is summarized in Figure 5.

2.1 Conceptual delineation of related constructs

In this paper, we distinguish three related but distinct terminologies: (1) Negative rejection, which refers to the model's inability to handle unsafe or irrelevant queries; (2) Hallucination mitigation, which refers to the reduction of unsupported or fabricated outputs outputs; and (3) factual grounding, which refers to the alignment of responses with verifiable external evidence. Hallucination mitigation and factual grounding both enhance the reliability of the facts, while negative rejection determines whether a response should be made. Therefore, these three structures are complementary: grounding and mitigation ensure that what is said is reliable, while rejection ensures that nothing should be said. Table 2 summarizes the different focuses and evaluations of each structure, providing a comparative basis for the subsequent analysis.

Table 1 Comparison of Strategies to Address Negative Rejection

Survey Works	Analyzes FT/PEFT	Utilizes RAG	Integrates RAFT	Focus on Negative Rejection	Mitigates Hallucination	Provides Unified 4-Dim Categorization	Proposes Rejection Benchmark
Wang et al. (2024a, b, c)	✓			✓	✓		
Cao (2024)	✓			✓	✓		
Zhu et al. (2025)	✓			✓	✓		
Farquhar et al. (2024)						✓	
Zhang et al. (2024a, 2024b, 2024c, 2024d, 2024e)		✓			✓		
Zhang et al. (2024b)	✓	✓	✓	✓	✓		
Dhuliawala et al. (2024)					✓	✓	
Our Work	✓	✓	✓	✓	✓	✓	✓

2.2 Literature search strategy

2.2.1 Keyword Selection

Keywords were iteratively refined to maximize recall and precision (see Fig. 4). Core search terms included "negative rejection in LLMs," "fine-tuning for rejection," "retrieval-augmented generation rejection," and "RAFT NLP." Auxiliary terms such as "LLM hallucination rejection" and "NLP misinformation mitigation" broadened scope. Boolean operators were applied (e.g., "negative rejection" AND "large language models"; "RAFT" OR "retrieval-augmented fine-tuning") to structure the search space.

2.2.2 Databases and sources

We targeted peer-reviewed venues and high-quality preprints, focusing on authoritative outlets such as NeurIPS, ACL, and ICLR. Searches spanned IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar. Additionally, public code repositories (e.g., GitHub) and benchmark descriptions were reviewed to capture implementation details and evaluation practices. While benchmark datasets (e.g., StackOverflow, RGB, HuggingFace listings) are discussed, no new datasets were created for this review.

2.3 Inclusion/exclusion criteria

A rigorous selection process was applied to ensure representativeness and scientific quality. Studies were included if they:

1. Directly examined negative rejection in LLMs.
2. Reported empirical results or conceptual frameworks related to fine-tuning, RAG, or RAFT.
3. Provided reproducible methods and standardized metrics (e.g., rejection rate, AUPR), as shown in Fig. 5.

2.4 Process

Exclusions were made for studies lacking methodological rigor, opinion-only pieces, or irrelevant domains. Priority was given to peer-reviewed publications from major conferences, while arXiv preprints were only included if subsequently accepted. The Prisma-style flow diagram (Figure 5) shows the number of retained and excluded studies at each stage.

2.5 Data extraction and synthesis

From each selected study, we extracted information on model architectures (fine-tuning methods, RAG mechanisms, RAFT strategies), dataset characteristics, evaluation metrics (e.g., rejection rate, false acceptance rate, F1-score, AUPR), and experimental results. Strengths, weaknesses, and limitations were also recorded. This structured process enabled systematic comparison across studies.

2.6 Citation tracking and snowballing

In addition to the database searches, citation tracking and snowballing were a part of the review process to inform systematic coverage of the literature. In the first step of citation tracking, we tracked new studies that had cited our included papers (i.e., forward citation), ensuring we captured the latest advancements. In the second step of citation tracking, we used backward citations to determine which studies were cited within the reference sections of our included articles, allowing us to identify studies that may have previously overlooked fundamental research (Badampudi et al. 2015). We used both Google Scholar and Semantic Scholar to undertake the citation impact analysis, favoring studies with the highest citation counts from the relevant domain.

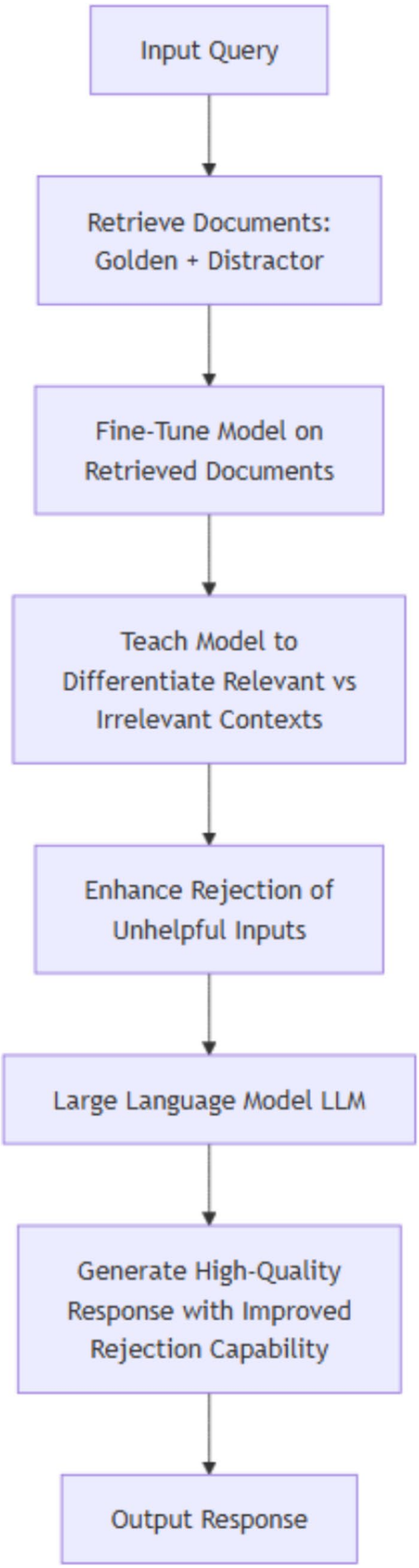

Fig. 3 RAFT implementation

Additionally, we explored the most current publication status for arXiv preprints to assess the credibility and scientific contribution of those papers.

2.7 Tools for analysis (i.e., Statistical/Visualization Tools)

Data handling and visualization were carried out using Pandas, NumPy, Scikit-learn, and Matplotlib, with supplementary visualizations created in Microsoft Excel, Visual Paradigm, and draw.io. These tools supported quantitative summaries and graphical illustrations of workflows.

3 Literature Review: Key Concepts and Approaches

3.1 Negative rejection in LLMs

3.1.1 Definition and context

Negative rejection refers to a language model's ability to abstain from responding to queries that are irrelevant, ambiguous, or outside its knowledge scope, thereby preventing misleading or incorrect outputs (Cao 2024; Chen et al. 2024). As illustrated in Figs. 6 and 7, this capability, distinct from hallucination reduction (mitigating confident but incorrect responses) and factual grounding (ensuring responses are verifiable), is critical in high-stakes domains like medicine, law, and education, where inappropriate responses can cause harm (Cao 2024). Unlike traditional metrics such as fluency or accuracy, negative rejection focuses on successful abstention, often implemented through refusal mechanisms like confidence thresholding or out-of-scope detection (Cao 2024). Many language models generate fluent but factually incorrect outputs due to a lack of robust refusal mechanisms, leading to hallucinations—confident but fabricated responses (Chen et al. 2024). This tendency to respond without sufficient information underscores the need for reliable negative rejection systems to enhance model trustworthiness and safety in critical applications (Cao 2024; Liu et al. 2025a, b).

This sharp shift in layer-wise dependencies suggests that monitoring the information flow from retrieved tokens to answer tokens could serve as an early signal for hallucination (Yu et al. 2024) and motivates the approach outlined in the Future Work part.

3.1.2 Impact on model performance

If there are no ways to correct their mistakes, LLMs can cause major harm in fields such as medicine, law and education. Mistakes in answering unusual medical problems in

Table 2 Comparative table of Negative Rejection, Hallucination Mitigation, and Factual Grounding

Construct	Core Question	Typical Metric	Example Benchmark
Negative Rejection	Should the model refuse?	Refusal rate, false rejection rate	SORRY Bench (Xie et al. 2024),
Hallucination Mitigation	Is the answer correct?	Hallucination rate, semantic entropy	RGB (Chen et al. 2024)
Factual Grounding	• Is the answer supported by evidence?	Evidence alignment, citation accuracy	FACTS Grounding (Jacovi et al. 2024) +

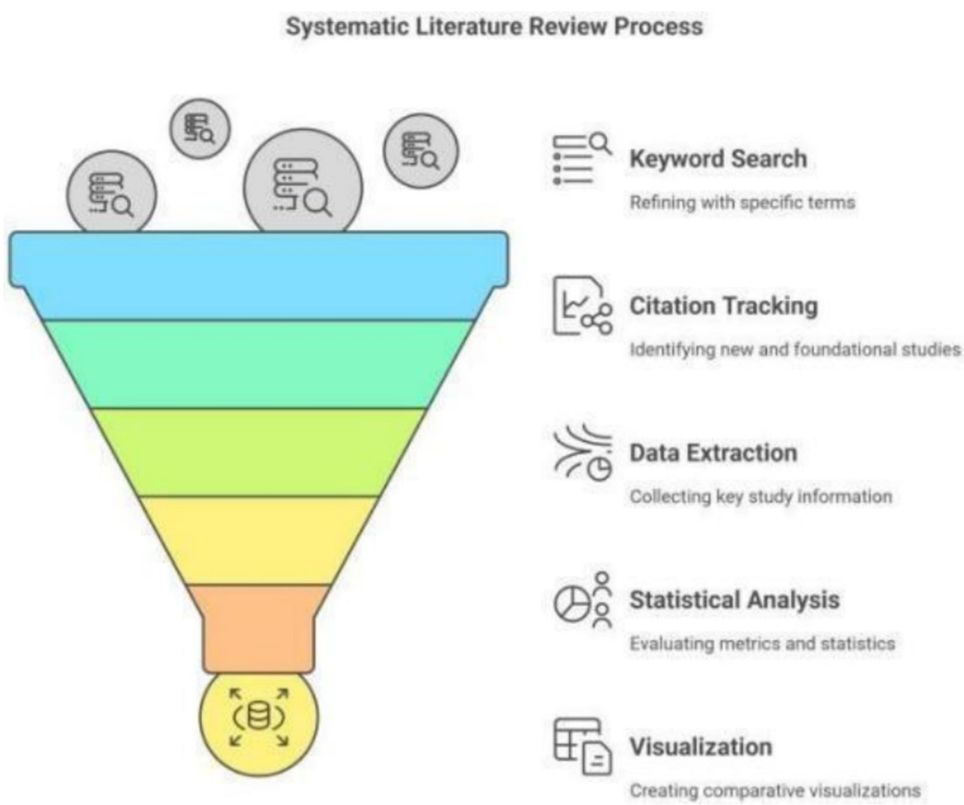

Fig. 4 Stages of the Systematic Literature Review (SLR)

healthcare have often resulted in incorrect diagnoses and accidents. Being overconfident about the incorrect answer in education may misdirect users and reduce how much they learn. Therefore, having systems that can understand their inability and properly decline to answer is very important (Feldman et al. 2024). When there is no rejection, users may start to question whether they can count on the AI in urgent situations like law advice, where mistakes can be very serious. Presence of rejection features helps lessen these risks and encourages LLMs for important decision-making purposes.

3.1.3 Measurement challenges

Traditional metrics like accuracy or precision fail to capture a model's ability to reject inappropriate queries, as they measure correctness but not abstention. Although rejection rates and false positives help, they do not consider the abilities of labels to resist malicious methods. The Retrieval-Augmented Generation Benchmark (RGB) addresses this by testing noise robustness and adversarial tasks to evaluate rejection capabilities (Chen et al. 2024). Currently, most methods use people to label data or work using given rules which limits how much they can expand. Automated evaluation frameworks using synthetic datasets with controlled noise levels are critical for comprehensive assessment (Bengio et al. 2025). Recent work on representation-space analysis reveals distinct rejection regions in models, enabling targeted editing to enhance refusal accuracy without compromising performance (Liu et al. 2025a, b).

3.1.4 State-of-the-art solutions

Addressing negative rejection involves implementing targeted strategies to improve a model's ability to abstain from answering irrelevant or unsupported queries:

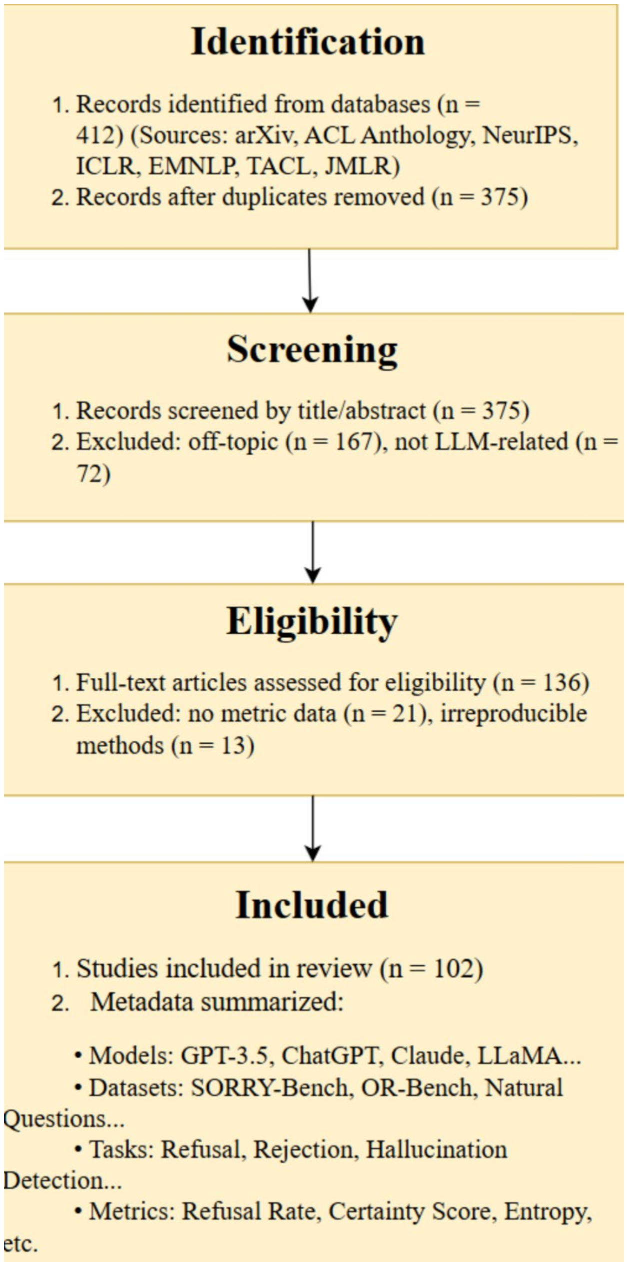

Fig. 5 Flow Diagram with Metadata Summary of Included Studies

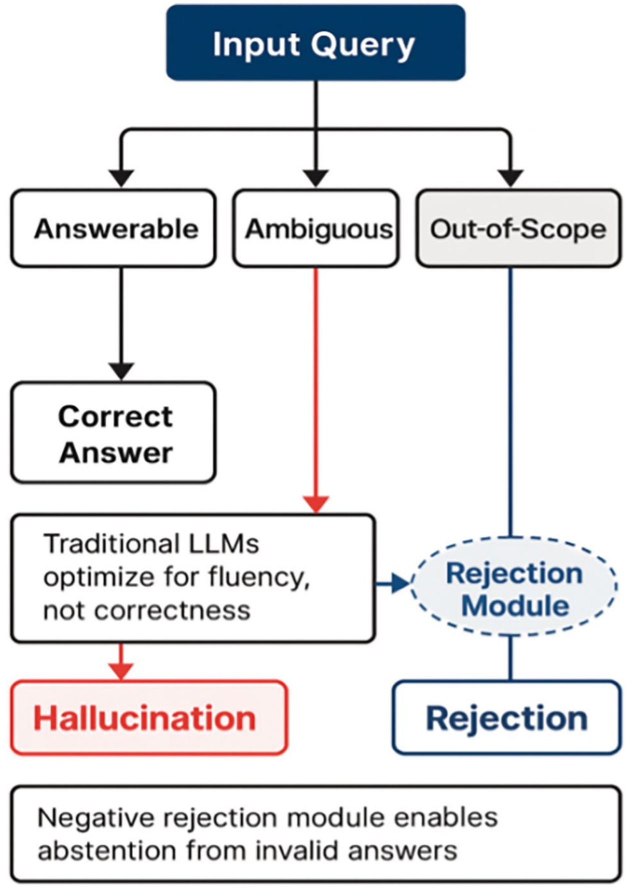

Fig. 6 Model response behaviors under uncertain or unsupported inputs

Fine-tuning on Specific Tasks: By using fine-tuning, models can improve their rejection skills using datasets of unanswerable questions. For instance, some datasets carry features that indicate to the model when it should not make a prediction. While effective in controlled environments, such models often struggle to generalize across diverse domains (Chen et al. 2024).

Incorporating the RAG: Incorporating RAG enables the use of external knowledge bases during answer generation, grounding evidence that might otherwise be hallucinated and therefore enhancing negative rejection when context is insufficient (Joren et al. 2025; Feldman et al. 2024). For technical queries, RAG can search corpus documents and use the retrieved context to guide responses. However, it relies on the external retrieval model to filter contextual noise (Feldman et al. 2024).

Retrieval Augmented Fine Tuning (RAFT): RAFT builds on RAG by integrating retrieval into fine-tuning, allowing models to better distinguish relevant from irrelevant context. Like a student learning to prioritize trustworthy sources, RAFT trains models to focus on meaningful content and ignore distractors. However, in practical deployment, several limitations arise, particularly related to computational cost and latency. Retrieval-based methods such as RAFT can significantly increase inference time, especially in real-time applications where rapid responses are crucial, such as in conversational AI or medical decision-making. Furthermore, the reliance on high-quality labeled datasets for fine-tuning and retrieval may result in domain-specific overfitting, limiting RAFT's generalization ability across various fields. These challenges must be addressed for RAFT to be effectively deployed in resource-constrained or dynamic environments (Oliveira et al. 2024).

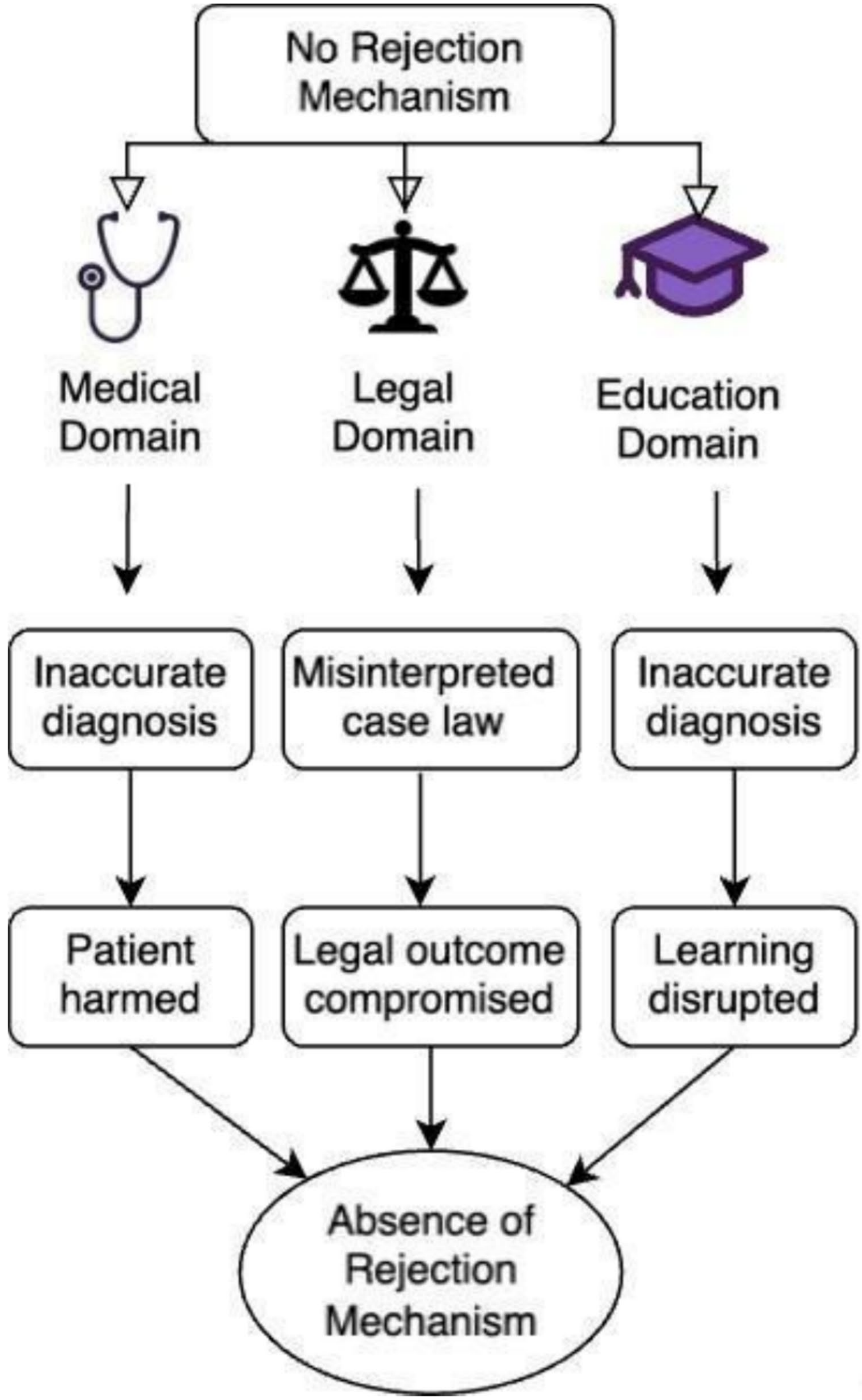

Fig. 7 Negative rejection failure across high-stakes domains

Techniques such as contrastive learning expose the model to conflicting data, teaching it to reject misleading inputs (Chen et al. 2024). While promising, RAFT's scalability is limited by its high computational and data requirements. Together, these approaches can enhance LLMs' ability to reject unreliable queries and improve their trustworthiness in real-world applications. Table 3

Table 3 Comparison of Strategies to Address Negative Rejection

Approach	Key Features	Strengths	Limitations
Fine-Tuning on Specific Tasks	• Uses labeled datasets with "no-answer" examples • Explicitly trains abstention behavior	• Effective in controlled environments • Directly teaches rejection criteria	• Poor generalization across domains
Incorporating Retrieval (RAG)	• Integrates external knowledge sources • Grounds responses in retrieved context	• Reduces hallucinations • Enhances contextual relevance	• Vulnerable to noisy retrieval outputs • Limited rejection capability improvement
RetrievalAugmented FineTuning (RAFT)	• Combines retrieval with fine-tuning • Uses contrastive learning and distractor documents	• Improves rejection ability • Robust to ambiguous/noisy inputs	• Computationally expensive • Requires high-quality labeled data

3.2 Fine-tuning for task adaptation

3.2.1 Mechanisms and workflows

Fine-tuning adapts pre-trained models to specific tasks using labeled datasets through a seven-stage pipeline critical for performance optimization (Parthasarathy et al. 2024). Data Preparation ensures dataset cleanliness and label consistency with pre-training vocabularies (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e). During Model Initialization, an architecture is picked that supports the task and in Training Environment Setup, resources and settings are arranged.

To preserve the knowledge learned before, Fine-Tuning uses supervised learning and methods like adapter layers, LoRA and freezing the weights of layers. Testing on data inside the domain and outside the domain is used for evaluation and validation. When the models are validated, Deployment is responsible for building the model into the production system and Monitoring allows for constant checking and model improvements.

With this method, improvement of LLMs for various purposes can be efficiently arranged in a repeated seven-stages sequence (Fig. 8).

Adapting LLMs seeks to ensure they do the best on each task with minimal costs and while keeping their general knowledge. According to Zhang et al., simple approaches include freezing part of the layers, boosting layers with text-specific tasks and regularizing the network by using dropout and weight decay to ensure no overfitting happens (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e). They simplify computation by using lower-dimensional representations which allows them to converge faster without losing their ability.

Using RLHF, evaluators' opinions are gathered to emphasize and ensure that the text is both safe and valuable. Optimizing CVaR helps reduce harmful results and does not affect the quality of a company's performance (Chaudhary et al. 2025). Together, these strategies enable efficient, generalizable fine-tuning across applications. Table 4

3.2.2 Role in enhancing negative rejection

Fine-tuning boosts negative rejection, the ability to abstain from irrelevant or unsupported queries, by embedding refusal mechanisms (Cao 2024; Chen et al. 2024). Unlike hallucination reduction or factual grounding, it prioritizes abstention via confidence thresholding and out-of-scope detection (Zhu et al. 2025). Refusal-aware instruction tuning trains models on unanswerable queries, rewarding refusals (e.g., "I don't know") for safety in medicine and law (Chaudhary et al. 2025). Adversarial training enhances robustness against ambiguous inputs, while instruction tuning embeds refusal protocols, addressing naive fine-tuning's limitations in high-stakes domains (Zhu et al. 2025). Table 5

3.2.3 Limitations in negative rejection ability

Fine-tuning boosts domain-specific accuracy with lower resource use than retraining (Chen et al. 2024). Naive fine-tuning risks overfitting, but S3FT balances specificity and generalization, cutting false rejection rates by 5.7% across four domains and boosting accuracy by 2.1% in GSM8K, 3.6% in MBPP, and 2.5% in Natural Questions (Gupta et al. 2025). Refusal-aware instruction tuning enhances negative rejection by training on unanswerable

Fig. 8 The seven stages of the fine-tuning LLMs

Table 4 Comparison of Different Fine-tuning Techniques in FineTuned LLMs

Fine-tuning Techniques	Methodology and Approach	Evaluation Metrics	Real-World Applications	Advantages	Limitations
Adapters (Hu et al. 2023)	Evaluates adapter placement, configuration, and performance; introduces the LLMAdapter framework	Accuracy, compared with GPT-3.5 and ChatGPT	Allows smaller models (e.g., LLaMA-13B) to outperform larger ones in specific tasks	Diminishes computational cost while reaching performance on par with full finetuning;decreases forgetting	Did not test larger models or consider adapter mixtures
Low-Rank Adaptation (LoRA) (Hu et al. 2022)	Decomposes weight matrices into low-rank matrices for adaptation	Performance on task-specific benchmarks, and memory usage	Large-scaleNLP tasks and domain adaptations	Efficient, converges quickly, and preserves the knowledge of the pre-trained model	May lead to lower performance on highly specialized tasks
QLoRA (Dettmers et al. 2023)	Extends LoRA by quantizing the weight matrices	Computational efficiency and task accuracy	Large model fine-tuning with limited resources	Reduces memory usage and computation in LoRA	Quantization may impact the model's accuracy
Weight-Decomposed LoRA (DoRA) (Liu et al. 2024)	Decomposes weights into smaller rank components and integrates them	Accuracy, computational efficiency, and performance	Large-scale models require memory-efficient fine-tuning in multitask settings	Combines LoRA's efficiency with better accuracy in parameter reductions	Complex to implement with greater computational overhead

Table 5 Comparison of Several Techniques to Improve Negative Rejection Ability in Fine-tuned LLM

Author (Year)	Measure	Methods	Benefits	Limitations	Datases
Wick et al. (2020)	Perplexity gap between positive and negative data	Anti-modeling using negative data to detect/attenuate undesirable statistical signals (e.g., n-grams)	Reduces overfitting to spurious n-gram signals; Improves syntactic learning	Complex optimization; May lead to over-penalizing legitimate patterns	Penn Tree Bank (PTB), subject-verb agreement tasks
Wang et al. (2024a, b, c) Task accuracy in reasoning/Q&A tasks	Negative-Aware Training (NAT): Integrating negative examples withexplicit success/failure indicators	Improves learning efficiency in data-scarce settings; Enhances reasoning performance	Similar focus on learning from negative data for performance gains	GSM8K, ASDiv, SVAMP, MultiArith, HotpotQA, StrategyQA	
Huang et al. (2024)	Harmful score / Fine-tune accuracy	Booster: Regularization during alignment to mitigate harmful fine-tuning perturbations	(20% reduction vs. baselines); Maintains task performance; Reduces harmful output	Requires pre-alignment data; Overhead in alignment-stage computation	BeaverTails dataset, SST2,AGNEWS, GSM8K, AlpacaEval

queries, improving abstention in high-stakes settings (Zhu et al. 2025). However, as shown in Fig. 9, Negative-Aware Training (NAT) and RLKF rely on scarce labeled datasets (Li et al. 2020). Their computational demands challenge resource-limited applications like edge computing (Liu et al. 2025a, b). Self-supervised and semi-supervised learning reduce labeled data needs, enhancing negative rejection scalability for dynamic environments (Chuang et al. 2020; Shi et al. 2023). Advanced methods ensure robust refusal, addressing naive fine-tuning's weaknesses (Cao 2024).

3.3 Retrieval-augmented generation (RAG)

3.3.1 Architecture and workflow

Lewis et al. explain that RAG mixes an LLM's generation skills with a retrieval system to boost what is relevant and accurate in answers (Lewis et al. 2020). Figure 10 shows that the retriever goes through various data sources (for example, research articles and domain data) and matches passages to the input question using dense vector encoding and similarity scoring (e.g., cosine similarity). The generator, which is most commonly a transformer-based LLM, combines both the prompt and the retrieved information to generate text, requiring less on fixed parameters and hallucination risks. As an illustration, RAG uses the latest IPCC reports based on recommendations to check if its background knowledge is up to date (Borgeaud et al. 2022). The use of this approach lets the model easily adapt to new information and is therefore helpful for open-domain QA as well as technical support.

3.3.2 Role in enhancing negative rejection

RAG promotes the negative rejection, or the power to not respond to irrelevant, ambiguous, or unwarranted requests, by basing its judgment on the knowledge possessed externally to determine the relevance of the queries before answering it (Cao 2024; Lewis et al. 2020). In contrast to the idea behind hallucination reduction, or its ability to eliminate confident but inaccurate outputs, or factual grounding, which prefers to ensure that the answers are consistent with the verifiable data, RAG promotes negative rejection by allowing models to decline an answer in the case of retrieved context that is inadequate or irrelevant (Cao 2024; Gao et al. 2023). In question answering in the medical context, to give an example, RAG still retrieves peer-reviewed research and will not respond when no relevant evidence is found, thereby preventing the possibility of providing wrong references (Edge et al. 2024). This active dependency on the outside world avoids the need of frequent retraining and having clear citation paths, but because of the possible spewing of poor-quality retrievals, a high-power indexing must be done to guarantee a reasonable negative rejection rate when it comes to high-stakes applications such as medicine and law (Lewis et al. 2020; Edge et al. 2024).

3.3.3 Practical limitations

As can be seen from Fig. 11, RAG cuts down on incorrect replies by supporting outputs with relevant proof collected from the site, nonetheless it encounters some major issues (Lewis et al. 2020). As found in ACL 2024, unsupervised improvements in RAG showed they could lower the false rejection rate caused by retrieval noise by 9% (Xu et al. 2024). The information from the retrieval system must

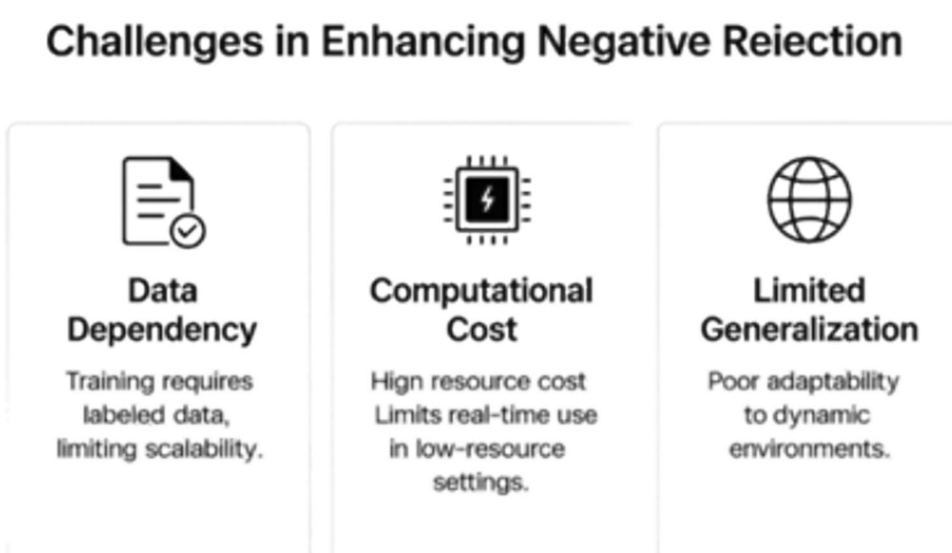

Fig. 9 Key Barriers to Enhancing Negative Rejection

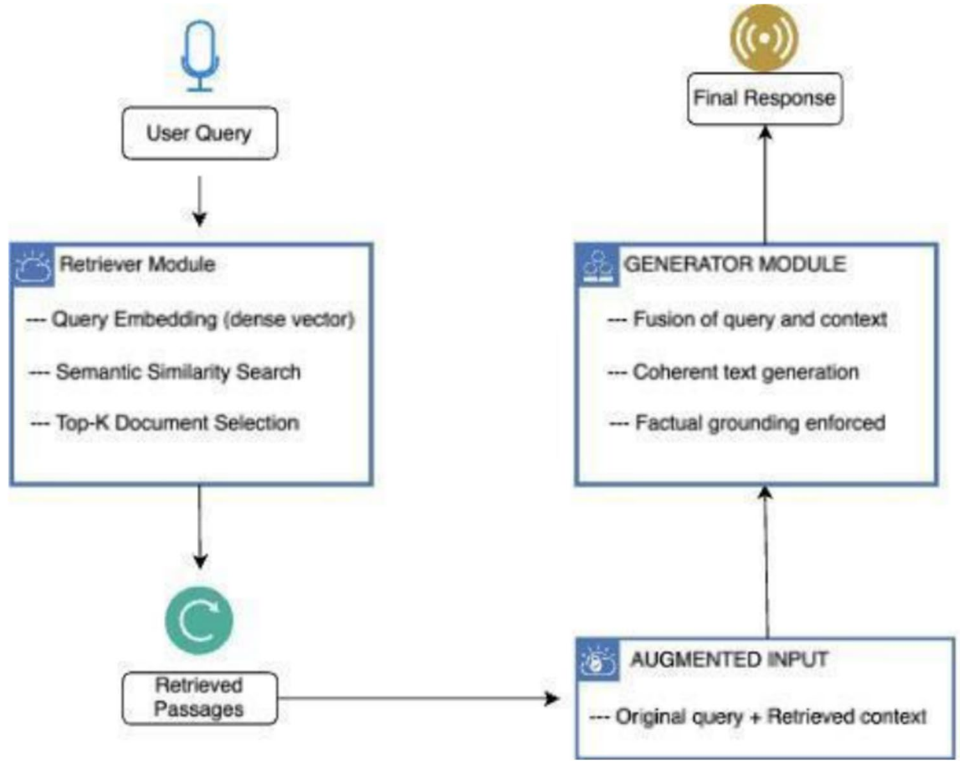

Fig. 10 Retrieval-Augmented Generation (RAG) architecture

be current and should not present conflicting facts if the LLM is to produce answers that are both right and easy to understand, especially in vital areas like medicine or law (Gao et al. 2023; Magesh et al. 2025). Figuring out whether documents contain true or misleading content can be difficult because RAG doesn't automatically review the documents (Edge et al. 2024; Gao et al. 2023). As an example, looking for previous cases might lead the model to pick unrelated evidence and come up with incorrect answers (Magesh et al. 2025). In essence, RAG's reliability depends on how well the retrieved content is of high quality and remains relevant (Lewis et al. 2020).

If retrieval does not guarantee context, the LLM may nonetheless produce faulty and confusing answers, which hinder its ability to eliminate incorrect claims. In fact, this is especially worrying in high-risk scenarios like medical and legal contexts where misinformation can be very costly (Magesh et al. 2025). The second is an issue of discrimination between informative and distractor documents (Edge et al. 2024). However, due to the fact that RAG does not automatically judge the trustworthiness and precision of retrieved documents (Fig. 12), it introduces the possibility of misinformation (Gao et al. 2023). For example, a retrieval system might conjure up conflicting or inconsistent case law, and the model might reach a similar conclusion, selecting an irrelevant or unhelpful evidence and returning a predicted false legal answer (Magesh et al. 2025). Additionally, RAG's reliance on external knowledge retrieval increases latency, as fetching and processing documents

Introduce additional computational overhead (Lewis et al. 2020). In real-time applications like conversational or predictive text applications, it is important to have a balance between retrieval speed, response speed, accuracy, and speed of response. To overcome these limitations, we need to improve document ranking, assess the source credibility, and filter retrieval documents such that RAG is able to reject the unreliable answers (Gao et al. 2023).

3.4 Retrieval-augmented fine-tuning (RAFT)

3.4.1 Synergy of RAG and RAFT

Building upon the idea of RAG, RAFT extends this paradigm by incorporating a retrieval mechanism in the fine-tuning to additionally improve a model's capability in distinguishing informative context and distracting context. During inference time (Fig. 13), instead of using a pre-trained model responsible for generating relevant documents in the RAG approach, RAFT uses retrieval awareness during training by training the model to learn what information is relevant and what information should be ignored (Zhang et al. 2024b). Another idea is M-RAG (Wang et al. 2024a, b, c), which significantly improves the accuracy rate of professional Q&A and simultaneously reduces false rejections through multi-partition retrieval (Wang et al. 2024a, b, c).In contrast, the

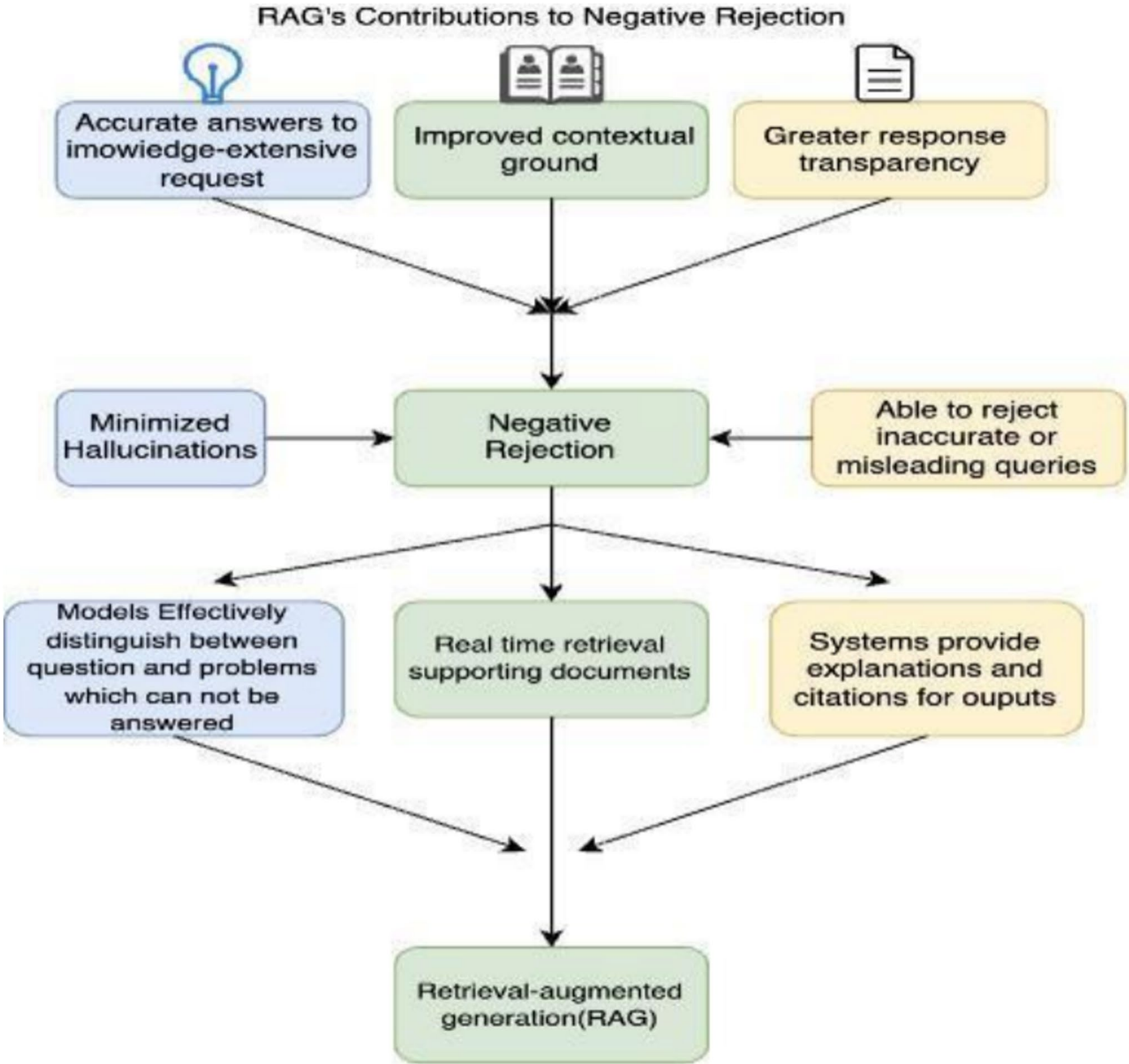

Fig. 11 Mechanism of RAG Integrating Retrieval for Accurate and Reliable Outputs

diap-and-then Align (DTA) method proposed by Sun et al. can automatically give a "I don't know" rejection answer when it detects that the query exceeds the dual knowledge boundaries of retrieval and the model, significantly reducing the error confidence rate (Sun et al. 2025). Following this training scheme, models create a more diverse understanding of retrieved content, allowing them to return responses in line with authoritative sources versus other generic pre-trained knowledge. Studies propose that RAFT is very beneficial in certain domain-specific tasks, for example, in the tasks of medical and legal question-answering, because of how models process and integrate retrieved information into responses (Warnakulasuriya & Hapuarachchi 2024).

It should be noted that although RAG enhances the factual basis by leveraging external evidence and indirectly reduces illusions, it will not fundamentally improve negative rejections unless combined with explicit rejection training. RAFT expands this paradigm by explicitly instructing the model to filter out irrelevant documents, thereby improving the factual basis and rejection behavior. In other words, RAG focuses on "better answering", while RAFT additionally trains the model to "know when not to answer".

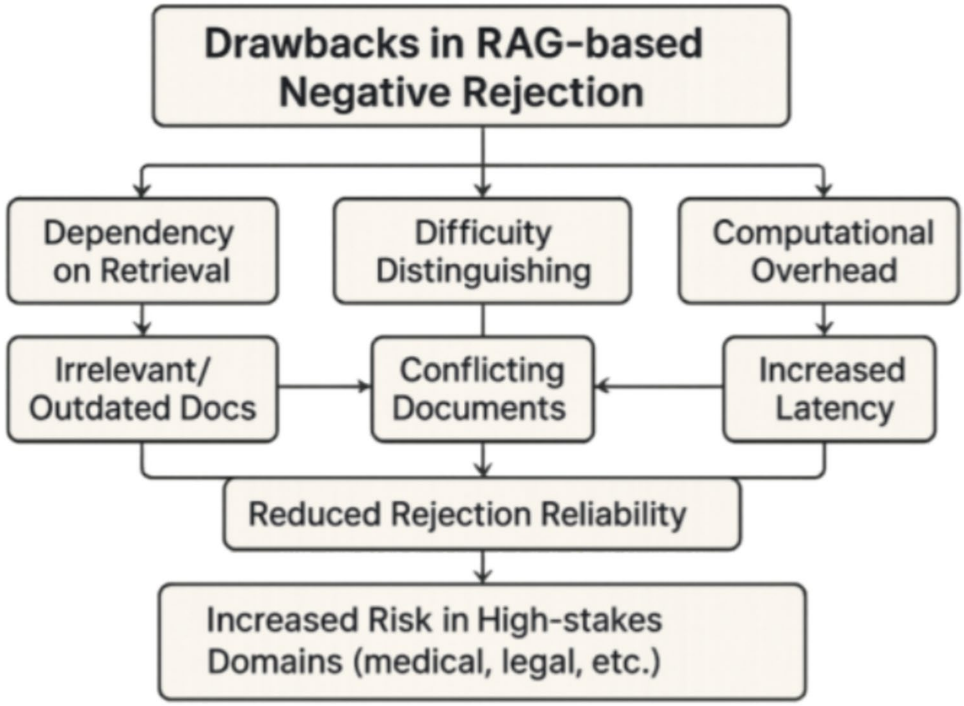

Fig. 12 Drawbacks in RAG-based Rejection Systems

3.4.2 Performance improvements

RAFT (Fig. 14) enhances negative rejection, abstaining from irrelevant queries, by integrating RAG's retrieval with fine-tuning (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e). Unlike RAG's 72% AUPR on RGB (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e), RAFT achieves 86.3% accuracy in rejection tasks (Oliveira et al. 2024). It ensures explicit refusals (e.g., "I don't know"). SLIM-RAFT's 8.63/10 accuracy in product classification beats Chat-GPT-4's 4.5/10, enhancing healthcare reliability (Oliveira et al. 2024). RAFT's hybrid design with rejection training justifies its distinct role.

3.4.3 Advances in RAFT and open challenges

One of the key innovations in RAFT is its use of any contrastive learning strategies to improve document selection (Fig. 15). In contrast with treating retrieved documents equally during training document relevance, RAFT fine-tunes models to give more weight to the retrieved evidence-based content by relevance and thus reduce dependence on retrieval noise (Zhang et al. 2024b). It lets us refine content selection in a model without introducing spurious marks into the code, as well as thinning low-quality evidence. In addition, RAFT enables a more effective chain of thought prompting during fine-tuning, yielding better reasoning ability in knowledge-based tasks. In other words, RAFT has improved the responses of models by requiring them to go through step-by-step reasoning with regard to retrieved evidence to produce reasonable and logically sound responses (Warnakulasuriya & Hapuarachchi 2024). In highly complex tasks such as medical decision making or legal analysis, reasoning and logical consistency are crucial, and this has been especially useful.

RAFT faces significant computational costs during retrieval-augmented fine-tuning, requiring exponentially more resources than traditional methods (Zhang et al. 2024b). Its reliance on labeled datasets—including annotated relevant and irrelevant documents—poses challenges for low-resource or multilingual contexts, where data scarcity limits scalability (Warnakulasuriya & Hapuarachchi 2024). For example, RAFT struggles in domains with insufficient high-quality evidence-based documents, hindering its effectiveness. Future work must address its dependency on extensive labeled data and expand applicability to low-resource settings.

Observed trade-offs in multilingual or low-resource settings. In languages with sparse labels and uneven corpus coverage, the retrieval side becomes a first-order bottleneck: dense retrievers show markedly lower effectiveness on Mr.TyDi and MIRACL languages compared with English, especially for morphologically rich scripts and code-switching queries (Zhang et al. 2021, 2023). In such conditions, RAFT tends to preserve its safety advantage (fewer unsupported answers) but often incurs higher spurious refusals and smaller end-task gains, because positive "relevant" evidence is scarcer and negative sampling is noisier. Cross-lingual open-retrieval settings—e.g., XOR-TyDi—further introduce translation/pivot steps that add latency and propagate

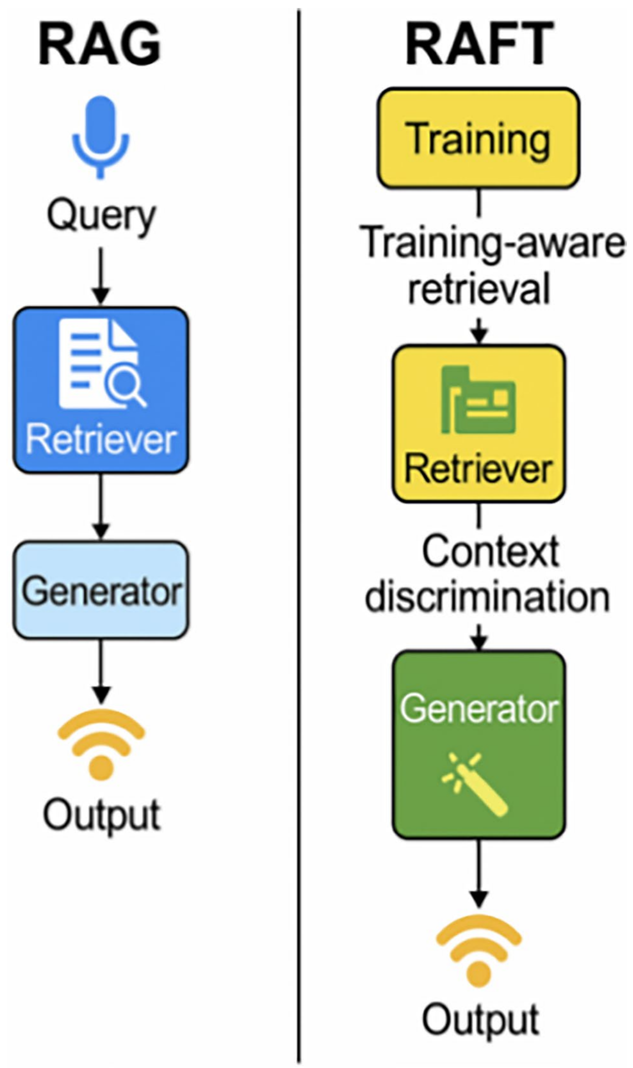

Fig. 13 Comparison between RAG and RAFT mechanisms

MT noise into the evidence pool (Asai et al. 2021); weak cross-lingual alignment can cause truly relevant passages to be treated as distractors during training, biasing the model toward abstention. Practically, this yields three coupled costs: (i) larger training footprints to construct bilingual relevant vs. distractor pairs, (ii) degraded retriever recall in low-resource languages (reducing RAFT's headroom over RAG), and (iii) a compute–latency trade-off where multilingual indexing and translation improve coverage but increase serving overhead (Clark et al. 2020; Zhang et al. 2021, 2023).

Brief example. Consider Swahili queries where answers reside mainly in English pages (an XOR-TyDi-style scenario): without strong cross-lingual retrieval/alignment, RAFT's training can under-expose true positives and over-expose mislabeled distractors, leading to conservative refusals under uncertainty; equipping the pipeline with multilingual dense retrievers and bilingual evidence mining narrows the gap but does not remove it when source corpora are intrinsically sparse (Asai et al. 2021; Zhang et al. 2021).

Implication. RAFT's headline benefits—robustness to retrieval noise and refusal rationales—remain, but the operating point shifts: in low-resource languages the model trades some utility (answers) for safety (refusals), and achieving English-like utility typically requires extra investment in corpus curation, multilingual indexing, and cross-lingual supervision (Clark et al. 2020; Asai et al. 2021; Izacard et al. 2021; Zhang et al. 2021, 2023).

4 Taxonomies and categorization

To analyze LLMs' negative rejection ability, we propose a four-dimensional framework organizing fine-tuning, RAG, and RAFT literature (Fig. 16). This scheme synthesizes existing studies across model architecture, evaluation metrics, application domains, and dataset characteristics. Architecture examines method-specific LLM modifications, metrics outline performance measures, domains assess applications in law, finance, and healthcare, and dataset characteristics evaluate data diversity's impact. This framework clarifies how each method influences LLM behavior, enhancing comparative understanding without introducing new evaluation tasks.

4.1 Model architectures

4.1.1 Fine-tuning

Traditional fine tuning can tune large amounts of model parameters to better adapt to the requirement of model for a specific task, however, it might cause overfitting problems. Parameter-Efficient Fine-Tuning techniques, such as Low-Rank Adaptation, LoRA, and Adapter Tuning, have gained huge public attention for many years. For example, LoRA was successfully used to fine-tune the ChatGLM3-6B-Base model on dataflow programming tasks in a study (Qiuming et al. 2024). The experimental results indicate that the BLEU score has increased from 0.091 to 0.193 using a growth rate of 112.1%, thus verifying the effectiveness of PEFT.

Fine-tuning also helps in handling the out-of-scope question queries and tweaks the model in such a way that unnecessary hallucinations do not take place. Recently, a study shows how fine-tuning can reduce hallucination in scientific text generation. The research team fine-tuned LoRA based on GPT-2 to improve performance and BLEU and Rouge scores, reducing the perplexity from 13,586.37 to 10,92.12 and simultaneously reducing illusions in scientific text generation (Sulimov 2024). AutoFT is an innovative auto-matic fine-tuning framework that automatically updates

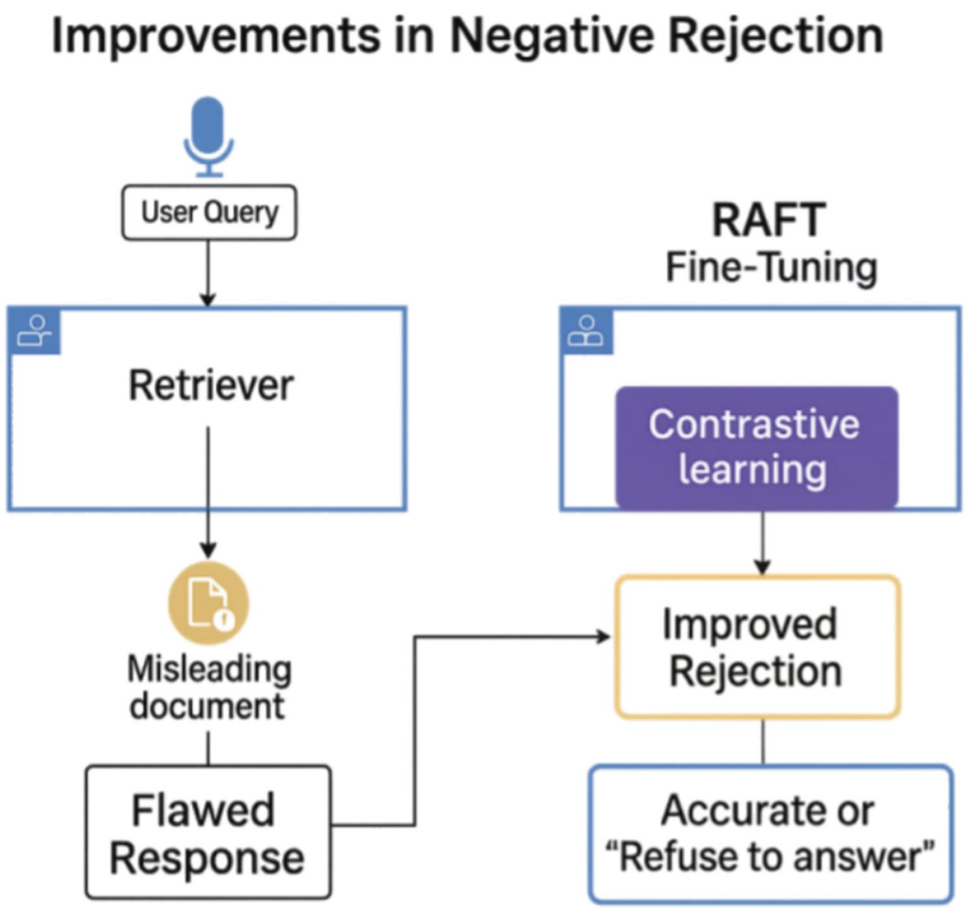

Fig. 14 RAFT's role in enhancing negative rejection by identifying unreliable retrieved content

the learning rate value of each model layer dynamically. The research of Choi et al. indicated that this method can bring a means to compensate for the collapse and catastrophic forgetting in a large amount of fine-tuning (Choi et al. 2024). Various NLP benchmark tasks (MNLI, QQP, SST, CoLA) were tested empirically, and it was verified that hierarchical learning rate optimization can improve task performance and generalization. Figure 17 compares the structures of three Fine-Tuning methods.

4.1.2 RAG

RAG combines the capabilities of external information retrieval and generation. Besides the data that was previously collected from the previous training, this tool is also capable of recovering the latest data from external resources. RAG internally stores information about all the contents of traditional fine-tuning methods and firstly fetches relevant information from massive databases or search engines, and combines it with context to generate answers. The retrieval and generation of an answer in this two-step process (Fig. 18) mostly makes the answer more accurate.

Initially, RAG diverts its attention to retrieving relevant knowledge fragments or documents from external sources. Then it puts together the output of the retrieved information fragments into the generative model. It introduces an extra level of contextual context for the model in order to enhance the quality of responses generated by the model in terms of, especially in open domain question answering and for document summarization. A recent research shows that RAG has large gains in factual grounding as well as contextual relevance on a variety of knowledge-intensive tasks. Results show that RAG architectures have lower hallucination rates compared to traditional LLMs, which makes them very fit for applications that require high factual accuracy (Tural et al. 2024).

As illustrated in Fig. 19, RAG greatly improves LLMs' capability of dealing with knowledge-demanding tasks by utilizing external information retrieval, but it poses new challenges, mainly for retrieval noise and accuracy of retrieval relevance. However, these limitations necessitate greater sophistication in the form of adversarial learning and chain of thought prompting that can be handled in larger frameworks such as RAFT.

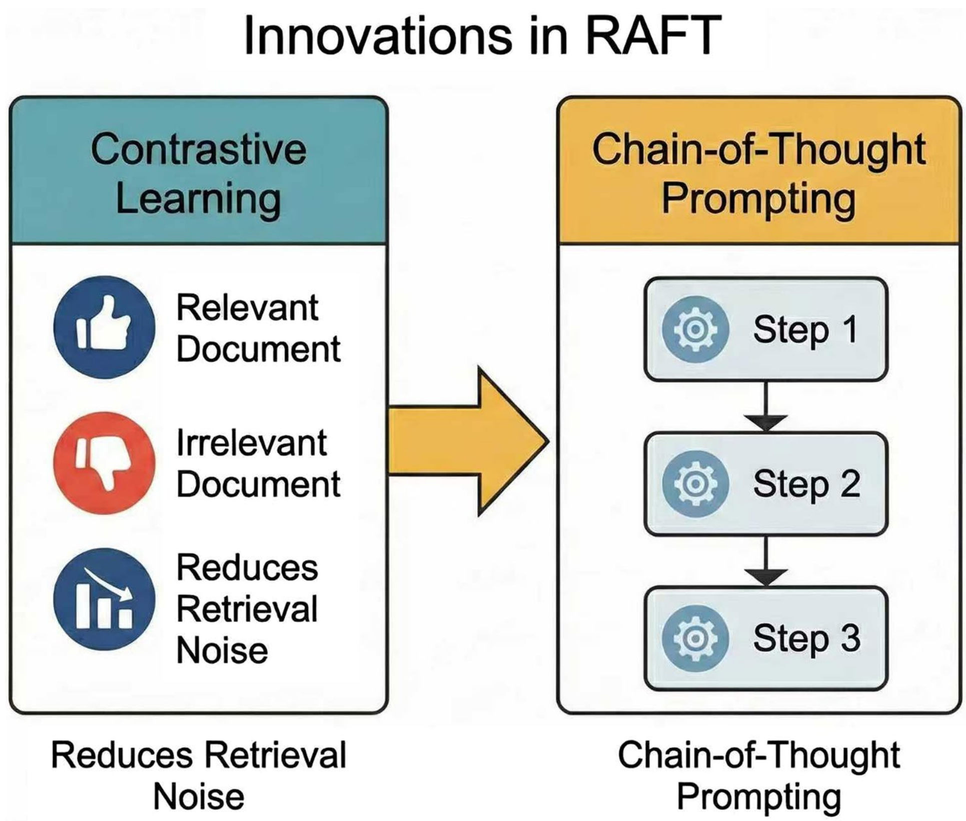

Fig. 15 Innovations in RAFT for Improving Negative Rejection

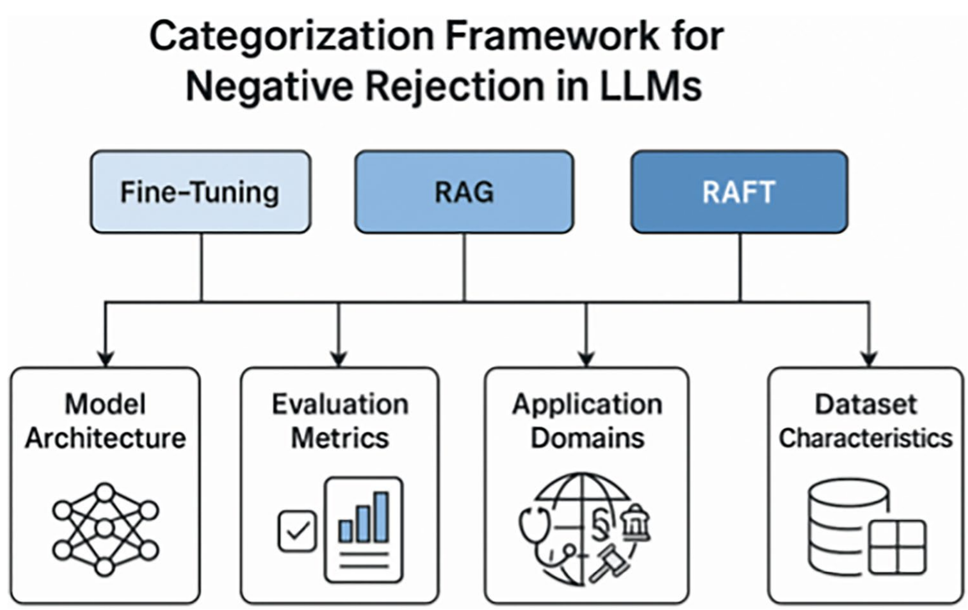

Fig. 16 A Unified Categorization Framework of LLM Negative Rejection Methods across Four Analytical Dimensions

4.1.3 RAFT

As shown in Fig. 20, RAFT extends RAG by combining retrieval with task-specific fine-tuning, enhancing robustness and contextuality. It trains models on relevant ("golden") and distractor ("irrelevant") documents, explicitly teaching them to distinguish useful inputs and reject noise (Oliveira et al. 2024). The SLIM-RAFT framework simplifies this approach, achieving 8.63/10 accuracy in Portuguese product classification vs. ChatGPT-4's 4.5/10,

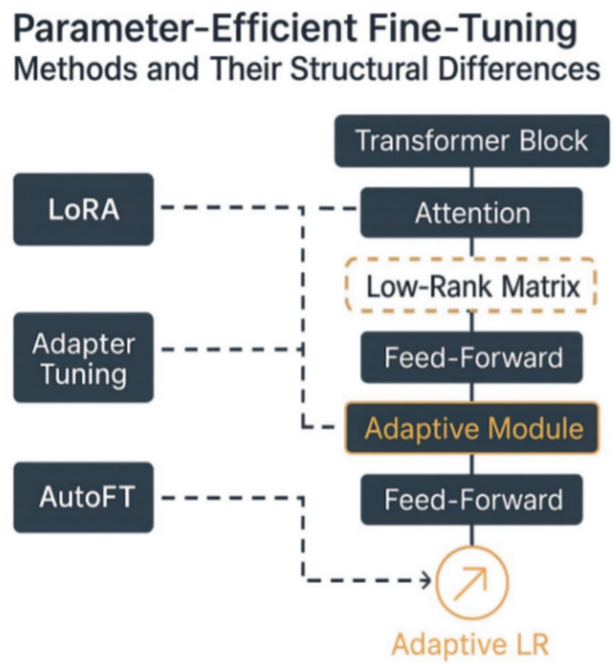

Fig. 17 Structural Comparison of Parameter-Efficient Fine-Tuning Methods

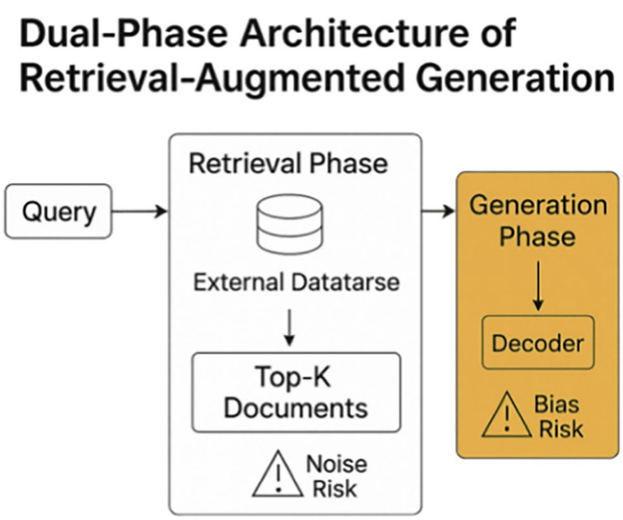

Fig. 18 Dual-Phase Architecture of Retrieval-Augmented Generation (RAG)

with lower computational costs (Oliveira et al. 2024; see Fig. 21).

RAFT integrates adversarial learning (exposing models to challenging inputs) and chain-of-thought prompting (step-by-step reasoning) to improve problem-solving and rejection of irrelevant content. This hybrid approach balances retrieval grounding with fine-tuned adaptability, addressing limitations of standalone RAG or fine-tuning methods

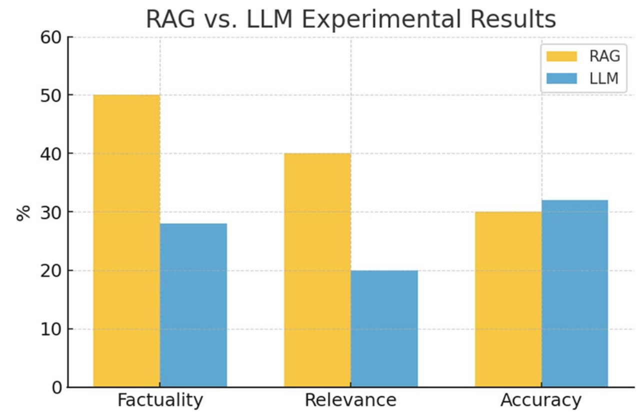

Fig. 19 Factual Accuracy and Hallucination Rate: RAG vs Standard LLMs

RAFT improves contextual relevance and reduces hallucinations compared to RAG or fine-tuning alone but faces limitations like high computational costs and sensitivity to noisy/outdated retrievals. Future work integrating parameter-efficient fine-tuning (PEFT) could enhance scalability and efficiency.

Stimulus RAG (SRAG) advances retrieval strategies by highlighting relevant sentences during prompting, significantly improving performance on low-frequency knowledge and long-tail entities while reducing hallucinations (Soudani et al. 2024).

4.2 Application domains

In recent years, the application of LLMs in particular industries, such as in the very high-risk areas of medicine, law, and finance, has been developed rapidly. For these fields, it is necessary to have accurate and reliable models that can resist fuzzy or meaningless input. RAG and RAFT have been proven to be very useful in these important fields by improving model performance and credibility.

4.2.1 Healthcare

The medical mining framework which is titled RAFT has been found to give satisfactory medical performance in terms of depth, medical relevance, and coverage of medical knowledge. Also, the RAFT system has another opportunity to diminish the amount of error outputs and promote the confidence of LLMs in diagnostic procedures that have high safety risks. On a more recent note, LLMs are normally being used in the medical field. They may help not only to write the medical record but also to be the helpful hints in the diagnosis process as well, up to the discovery of new medication. To minimize the hallucinations during medical Q&A, the EvoLLMs framework (Fig. 22) a combination of

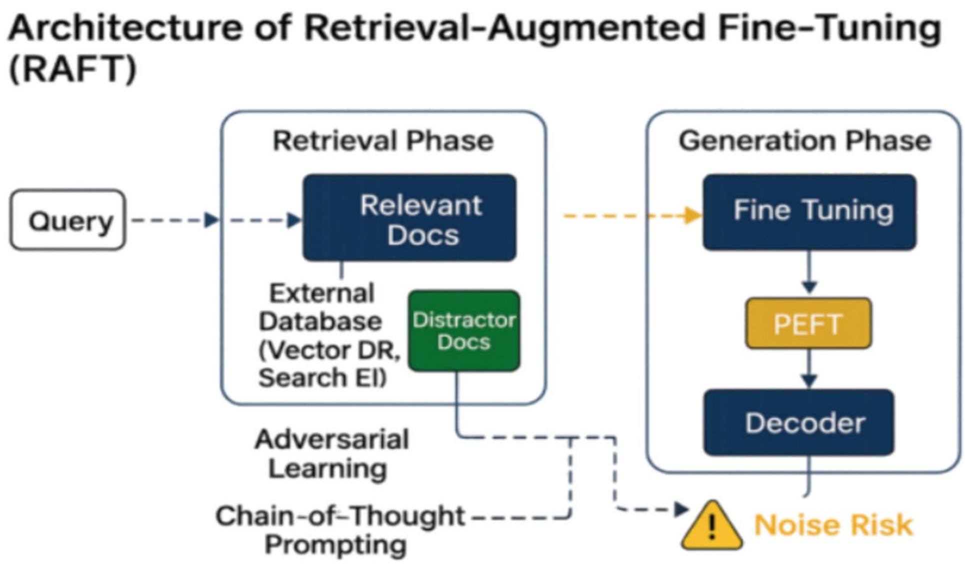

Fig. 20 Architecture of RAFT

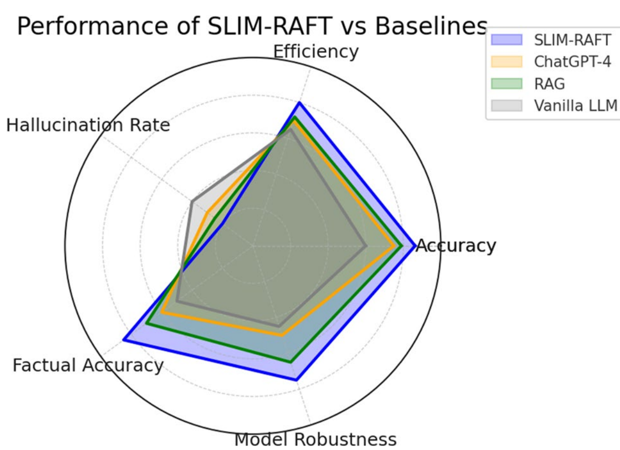

Fig. 21 Performance of SLIM-RAFT Compared to Baseline LLMs

genetic algorithms and LLMs was employed (Boulesnane and Souilah 2024). The outcomes of the present study demonstrate that such an enhancement would reduce hallucinations drastically even with severe medical issues. The models are very quantitatively comparable to human experts in the task in terms of depth, relevance and overall medical knowledge. Moreover, the error rate in RAFT is also low, which gives us the opportunity to assign the diagnosis of shortcomings to LLMs to a certain degree more confidently.

4.2.2 Legal

These models are commonly used in the legal domain, where they are employed in inspecting contracts, organizing legal documents, and ensuring compliance with regulations, among others (Fig. 23). An experimental study presented mashed up In–Context Learning and Random Access Generator to sort out legal texts involving disputes over construction delays (Huber-Fliflet et al. 2024). Generally speaking, combining RAG with a few-shot learning approach bumped up both the accuracy and speed of document classification, which really helps when you're faced with a mountain of paperwork. The system identifies key legal provisions and maintains high accuracy and recall rates. Then the RAFT framework was applied to alleviate the hallucinations that occurred in legal reasoning, which helped to synchronize the output with the actual wording of real legal precedents and regulations. All in all, the integrated model will eventually bring more trust to the automated legal system.

4.2.3 Finance

In the financial field, LLMS have become important tools, which are usually responsible for assessing risks, detecting fraud, and predicting market changes (Fig. 24). A recent study explored the use of LLMs to assist in anti-money laundering and compliance risk prevention. They proposed a framework that combines LLM with an anomaly detection system to identify suspicious transactions. The results show that the recall rate has soared to nearly 97.44%, and the false alarms have decreased significantly. Also, integrating the RAG and RAFT frame- work can obtain external financial real-time data to facilitate the decision-making process (Mhammad et al. 2023).

4.3 Dataset characteristics

It's clear that the quality, variety, and focus of your datasets set the stage for how well large language models perform. These factors have a huge impact on whether methods like Fine-Tuning, RAG, and RAFT can adhere to facts, reduce

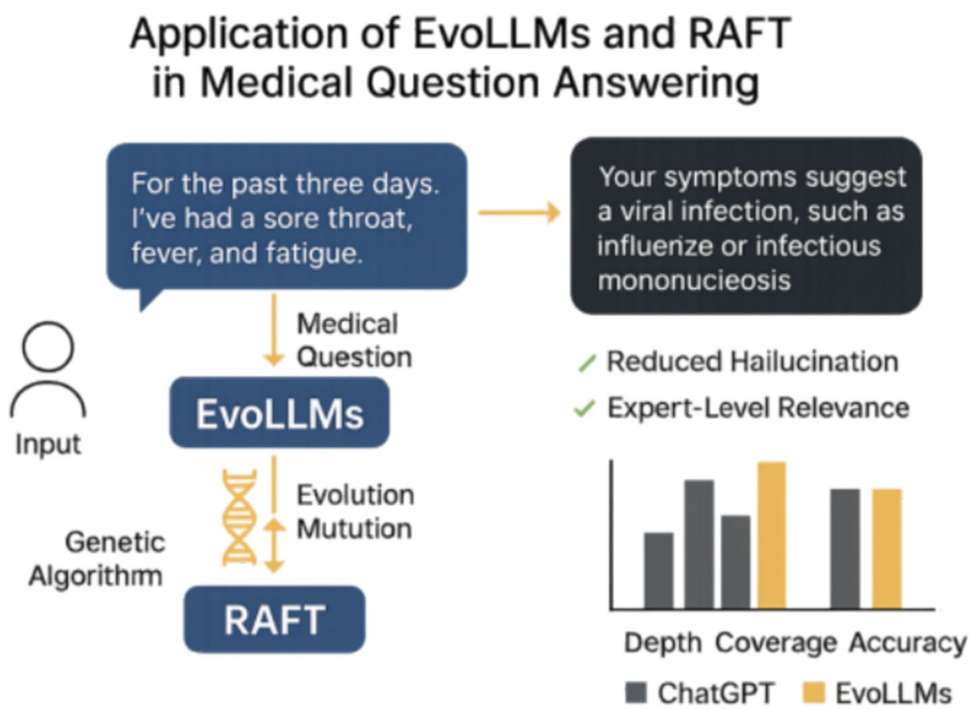

Fig. 22 Application of EvoLLMs and RAFT in Medical Question Answering

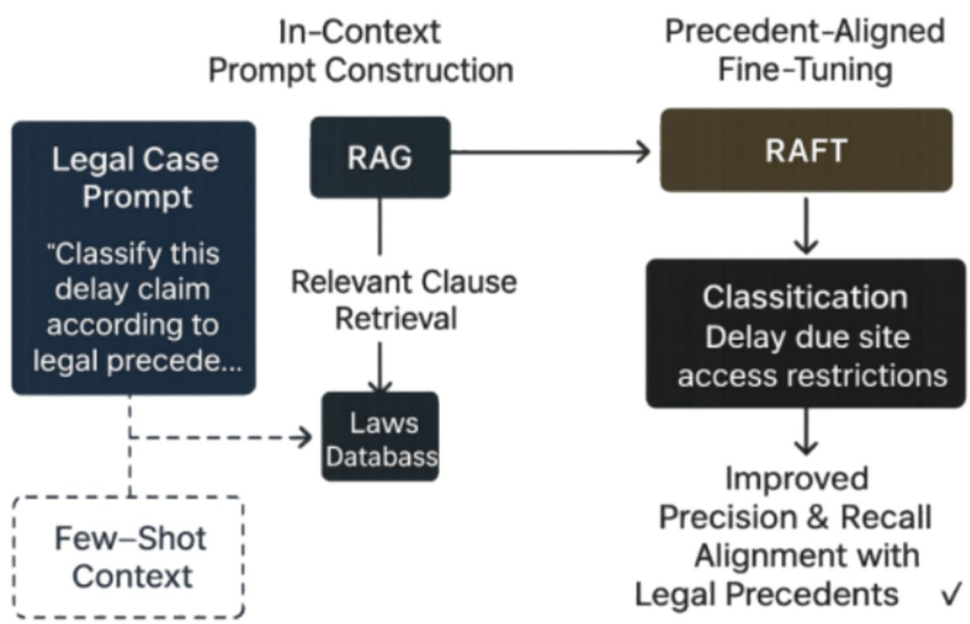

Fig. 23 RAG and RAFT-Enhanced Legal Document Classification with In-Context Learning

made-up details, and disregard fuzziness or inputs that miss the mark. The general theme of well-organized dataset can really tune model outputs and keep them stable when the tasks are mixed is hammered in recent works.

4.3.1 Data quality and relevance

High-quality datasets are required to prevent hallucinations and increase factual consistency in the output from LLMs. In the matter of fact, Li et al. presented an adaptation of In-Context Learning (ICL) within a Maximal Marginal Relevance (MMR) framework for the text-to-SQL task. They concluded that balancing relevant examples with diverse ones in a dataset functions greatly in complex SQL generation tasks, in terms of model performance (Li et al. 2024b). Applying the model using well-chosen high-quality data samples worked better in terms of the execution accuracy perspective as the model was capable of detecting whether data was related to itself, or if it was complete and did not tend to task-specific hallucinations.

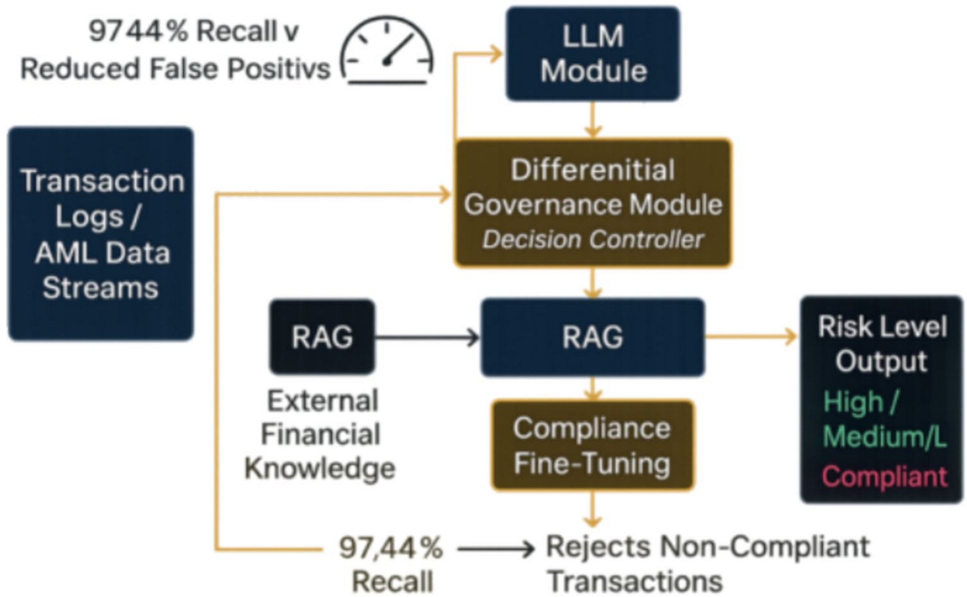

Fig. 24 LLM-Based Framework for Financial Risk Detection Using RAG and RAFT

4.3.2 Diversity and balance

For LLMs to possess generalization ability, it has to have data diversity, i.e., different language patterns, different domains, and different contexts to train models to accept fuzzy input. Jayawardena & Yapa mentioned that Graph-Based Retrieval Augmented Generation (G-RAG) could be employed to keep semantic diversity and entity coverage held in the task of synthesizing synonyms (Jayawardena & Yapa 2024). Graph-based knowledge retrieval methods can result in great improvements in semantic similarity, lexical diversity, as well as syntactic variation. Furthermore, results depict that the LLMs can achieve a good understanding and generation of correct text in different contexts in many fields with the help of diverse datasets. Data balance can help reduce model bias and improve generalization. Jayawardena & Yapa emphasized that integrating knowledge graphs in G-RAG helps ensure that the involved datasets contain a large number of different entities and relationships, thereby providing a better factual basis for LLMs' output (Jayawardena & Yapa 2024). In addition, it balances the data distribution, so the model does not overfit any specific pattern or category, and at the same time it can handle various given inputs. We summarized these in Fig. 25.

5 Methodologies and evaluation

5.1 Evaluation metrics

Evaluation Metrics are important tools for determining the efficiency of LLMS in negative rejection and reducing hallucinations. Robustness of LLMs has also been more recently researched by assessing their factual consistency, their capability of estimating uncertainty, and the capability of detecting hallucinations in various contexts. Farquhar et al. and Thomas et al. have recently published two studies that give thorough frameworks for evaluating the promise of those capabilities in LLMs (Farquhar et al. 2024; Thomas et al. 2024).

The negative rejection evaluation model lacks the ability to avoid answering irrelevant or unsafe queries, and focuses on when the model should reject. Unlike hallucination mitigation (detecting confident but incorrect outputs), negative rejection measures whether the system rejects unsupported requests, rather than fabricating an answer.

Key metrics include:

1. Refusal Rate: the percentage of harmful or out-of-scope queries correctly rejected.
2. False Rejection Rate: the proportion of valid queries wrongly refused.
3. Selective Accuracy: the combined ability to both answer in-domain queries and abstain from unsafe inputs.

Benchmarks targeting these metrics include:

• SafetyBench: Evaluates LLM safety across 14 dimensions (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e).
• OR-Bench: Diagnoses false rejections with 80,000 prompts (Cui et al. 2025).
• SORRY Bench: Analyzes rejection across 44 sensitive topics (Xie et al. 2025).

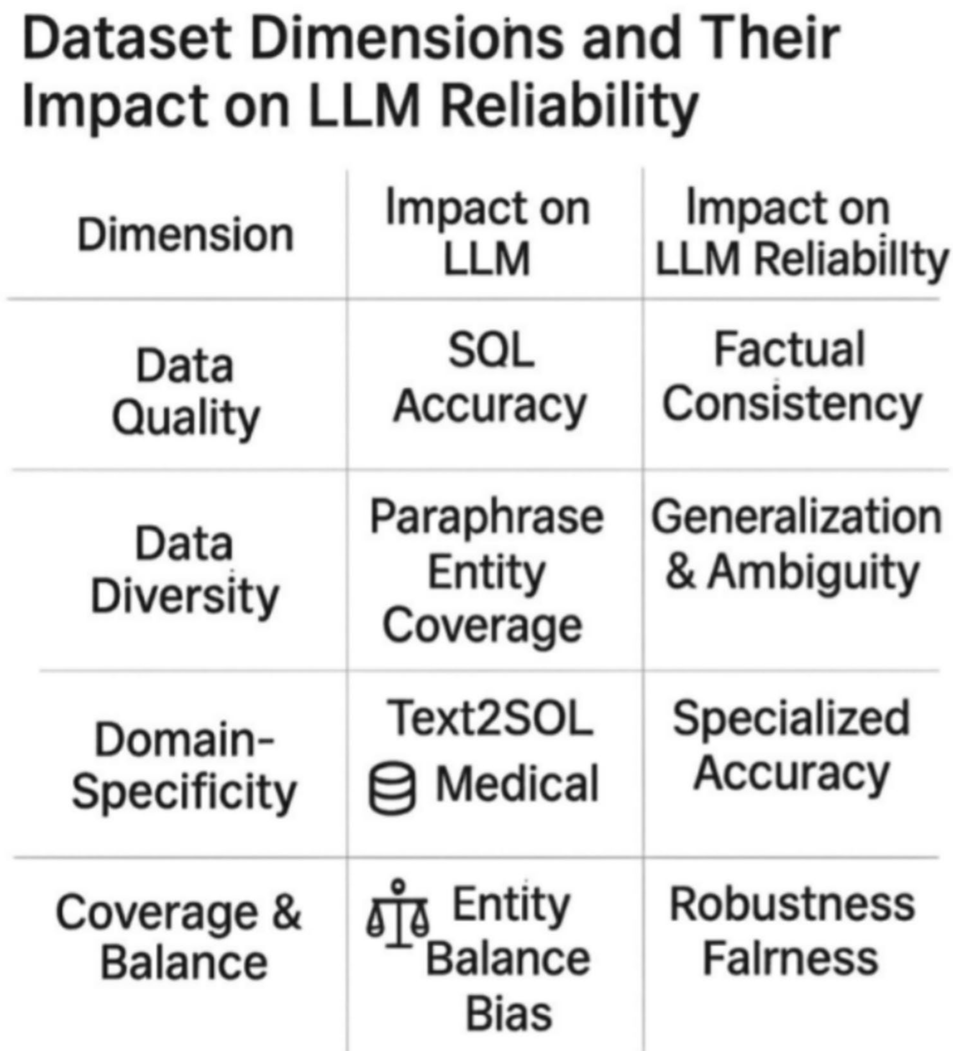

Fig. 25 Dataset Dimensions and Their Effects on LLMs

5.1.1 Task-Specific Benchmarks

Thomas et al. demonstrated LLMs' ability to evaluate human tutor responses using metrics like (Thomas et al. 2024):

- F1 Score: 0.85 (prediction), 0.83 (explanation).
- Cohen's Kappa: 0.69–0.65 (human-AI agreement).
- Pretest-Posttest Gains: Validated learning improvements.

The study highlights prompt engineering and Chain-of-Thought (CoT) prompting as effective tools for reducing hallucinations and improving reliability. Results emphasize the need for multi-dimensional evaluation, combining semantic entropy, accuracy, and hallucination detection frameworks. For instance, LLMs accurately assessed mentor responses in educational contexts, suggesting future applications in diverse fields (Thomas et al. 2024).

5.2 Performance Analysis

5.2.1 Quantitative comparisons (Accuracy, Efficiency)

This study does not present new experimental results. Instead, we conducted a comparative review of quantitative findings reported in prior research. Specifically, we examine metrics such as accuracy, rejection rate, computational efficiency, and hallucination frequency as reported in peer-reviewed publications and open-source benchmarks. This comparison aims to highlight relative strengths and weaknesses across methods like fine-tuned LLMs, RAG, and RAFT, focusing on their reported performance under similar evaluation settings.

Table 6 AUPR Comparison Across Fine-Tuning, RAG, and RAFT

Approach	AUPR (%)	Dataset/Task	Source
RAFT	79.0	RGB	(Zhang et al. 2024b)
RAG	72.0	RGB	(Zhang et al. 2024b)
Fine-Tuning	3.22 (Improvement)	StackOverflow	(Zhang et al. 2024a)

5.2.2 AUPR comparison

Table 6 compares the AUPR performance of Fine-Tuning, RAG, and RAFT, sorted by descending AUPR where applicable.

As the table demonstrates, RAFT achieves the highest AUPR of 79.0% on the RGB dataset, outperforming RAG's 72.0% on the same benchmark. Fine-Tuning demonstrates a 3.22% AUPR improvement on StackOver-flow, reflecting enhanced out-of-scope detection but not directly comparable due to differing datasets (Zhang et al. 2024a, b).

5.2.3 Accuracy comparison

Table 7 presents the Accuracy performance, with estimates for Fine-Tuning and RAG to ensure a comprehensive comparison.

RAFT leads with an accuracy of 86.3% in product classification, followed by Fine-Tuning with an estimated accuracy of 80.0% on StackOverflow and RAG with an estimated accuracy of 75.0% on RGB. These estimates are derived from AUPR performance and task-specific improvements (Oliveira et al. 2024; Zhang et al. 2024a, b).

5.2.4 Data sources and estimation methodology

The AUPR values for RAFT (79.0%) and RAG (72.0%) are directly sourced from Zhang, Patil, et al. on the RGB dataset, enabling a direct comparison. Fine-Tuning's 3.22% AUPR improvement on StackOverflow is from Zhang, Norouzian, et al., reflecting a relative gain in out-of-scope detection. For Accuracy, RAFT's 86.3% in product classification is directly from Oliveira et al. (2024). Due to the absence of absolute accuracy metrics for Fine-Tuning and RAG in comparable tasks, we estimated these values to facilitate comparison. Fine-Tuning's estimated 80.0% accuracy on StackOverflow is inferred from its 3.22% AUPR improvement (Zhang et al. 2024a), assuming a baseline AUPR of approximately 76.78% improved to 80%, with accuracy closely aligned due to the correlation between AUPR and classification performance in intent detection tasks. RAG's estimated 75.0% accuracy on RGB is based on its 72.0% AUPR (Zhang et al. 2024a), assuming a slight increase, as accuracy typically exceeds AUPR in retrieval-augmented settings. These estimates ensure at least two elements per table, as required, but highlight the need for standardized metrics in future studies.

5.2.5 Synthesized comparative evaluation

To provide a unified, metric-consistent comparison, we synthesize rejection performance, hallucination mitigation/factual grounding, and efficiency/deployability into a single ordinal (1–5; higher is better) heatmap/ranking table (Table 8) across Fine-Tuning, RAG, and RAFT. Unlike prior scattered tables (e.g., AUPR and accuracy listed separately), Table 8 provides a single, normalized view of trade-offs

Table 7 Accuracy Comparison Across Fine-Tuning, RAG, and RAFT

Approach	Accuracy (%)	Dataset/Task	Source
RAFT	86.3	Product Classification	(Oliveira et al. 2024)
RAG	80.0 (Estimated)	StackOverflow	(Zhang et al. 2024b)
Fine-Tuning	75.0 (Estimated)	RGB	(Zhang et al. 2024a)

Table 8 Synthesized comparative evaluation (1–5, higher is better)

Dimension	Fine-Tuning	RAG	RAFT
Rejection ability (AUPR & refusal)	2	4	5
Hallucination mitigation	2	4	4
Factual grounding	2	5	4
Training cost	5	2	3
Inference latency	1	4	2
Update flexibility (data refresh without retrain)	1	5	3
Domain adaptability / generalization	3	4	5
Labeled-data requirement	3	1	4
Robustness to retrieval noise/ambiguity	3	2	5

across FT, RAG, and RAFT, enabling safety-, latency-, or adaptability-oriented selections at a glance

Takeaways

• RAFT ranks highest for safety/robustness (rejection, noisy inputs).
• FT offers the lowest inference latency for real-time settings.
• RAG provides the best update flexibility for fast-changing knowledge.

Evidence basis

• Performance of rejection: The AUPR metric of RAFT on the RGB dataset is higher than that of RAG (79.0% vs. 72.0%), but the increase in AUPR of the fine-tuned results on another dataset is smaller; therefore, in terms of rejection operations, the performance of RAFT is better than RAG, and RAG is better than the fine-tuned model (i.e., RAFT > RAG > FT).
• Accuracy (representativeness): The accuracy rate of RAFT in product classification is 86.3%. The RAG/FT values here are not completely comparable across different tasks. These values are only listed in the paper as estimated values.
• Hallucinations and Stability: RAG reduces hallucinations through external evidence, but is susceptible to retrieval noise; RAFT can enhance adaptability to noisy/fuzzy inputs; while FT is more constrained by the limitations of static knowledge.
• Efficiency trade-off (training cost / inference latency / update flexibility): The training cost of FT is relatively high, but the inference latency is low; RAG has a moderate training cost, but the inference latency is high and the update flexibility is poor; RAFT has a moderately high training cost, with the inference latency ranging from low to medium, and the update flexibility is moderate.
• Domain adaptability and resistance to retrieval noise: RAFT performs more robustly in noisy/professional scenarios; RAG is adaptable but dependent on retrieval quality; FT is effective in static domains but has weak generalization ability.

5.2.6 Qualitative Insights

We conducted a qualitative analysis of each method's failure modes and strengths by reviewing 50 incorrectly answered or refused outputs per approach. These were labeled using an error taxonomy: hallucination (confident fabrications), spurious rejection (suppressed correct answers), and over-confident acceptance (incorrect answers without uncertainty). Fine-tuned models exhibited frequent hallucinations, RAG systems struggled with irrelevant/noisy retrievals, and RAFT models showed fewer errors due to adversarial training exposure.

Two domain experts (legal/financial) evaluated 20 outputs each, rating clarity, correctness, and trustworthiness (1–5 scale). RAG excelled when retrieval was accurate but faltered with irrelevant documents, while RAFT balanced precision and caution—often including refusal rationales when answers were unreliable. Expert feedback and error patterns jointly highlighted RAFT's robustness in high-stakes scenarios.

6 Comparative analysis

6.1 Strengths and weaknesses

This section compares fine-tuning, RAG, and RAFT for negative rejection (abstaining from irrelevant queries), efficiency, and adaptability. Table 8 summarizes these. RAFT achieves 86.3% accuracy (Oliveira et al. 2024), RAG 72% AUPR on RGB (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e), and S3FT 5.7% false rejection reduction (Gupta et al. 2025). Strengths are detailed below.

6.1.1 Negative rejection ability

Fine-tuning Models often have difficulty with rejecting irrelevant or out-of-scope queries because they only rely on the information incorporated in the model parameters. Fail to reject responding to unanswerable queries can lead to hallucinations when the model is dealing with ambiguous input (Ovadia et al. 2024). RAG is clear in improving the negative rejection ability in that the generated answers are grounded in retrieved documents. However, as RAG is reliant on the quality of the retrieval mechanism, the generated answer may still be inaccurate if the retriever returns noisy data (Fang et al. 2024a, b; Shuster et al. 2021). RAFT builds from these approaches by employing retrieval awareness, or awareness of retriever results, during the fine-tuning phase and by explicitly training the model distinguish actual relevant documents and distractors, improving the models rejection ability (Sager et al. 2024).

6.1.2 Scalability and computational cost

Fine-tuning incorporates reasonably detailed domain-specific knowledge but requires extensive computational resources during training, as it needs to update all parameters of the model, which is costly in terms of training time. Fine-tuning models can have low latency at the processing stages, but as stated, they lack the scalability in time because of high retrain cost (Ovadia et al. 2024). RAG improves efficiency by decoupling the retrieval process from model training, but it increases inference latencies from real-time document retrieval, which are much dependent on prior access to the properties and documents records provided by the retrieval method (Rao 2024). RAFT creates a balance between these approaches: although its training phase is more resource-intensive than RAG, the resulting model actually benefits from reduced inference latency by pre-learning how to integrate retrieved information effectively (Greyling 2024). As Fig. 9 illustrates, Fine-tuning suffers from long training time but have lower computational inference latency. RAG was shorter in training time but has a large overhead cost from inference approach latency, and RAFT combines moderate training duration with reduced inference latency, which effectively integrates domain-specific fine-tuning with dynamic retrieval.

6.1.3 Domain adaptability

Fine-tuning is primarily effective when it has separated sufficient detail in the domain to fine-tune it, and the model will adjust to an area, though it is static and does not change to new or evolving information (Zhang et al. 2024a). RAG generally performs adaptable, but its performance was purely quality based on the retrieval quality (Fang et al. 2024a, b; Izacard et al. 2022). RAFT combines the advantages of Fine-tuning and RAG by integrating domain-specific fine-tuning with the flexibility of retrieval, meaning it is able to perform well even under noisy and evolving conditions, not just adapt but likely perform better for specialized applications (Sager et al. 2024).

6.1.4 Practical deployment considerations

In summary, the deployment context significantly affects the applicability of RAG, RAFT, and Fine-Tuning, even though each has unique advantages. Because Fine-Tuning has a high inference speed after training but a low capacity to adapt to new information, it is better suited for situations where knowledge is relatively stable and latency is strictly required. Because RAG can call external information to ensure the promptness and accuracy of the answers, it has greater advantages in dynamic or knowledge-intensive scenarios. However, it also introduces retrieval latency and is susceptible to being affected by noisy documents. RAFT demonstrates stronger rejection and robustness through the introduction of retrieval awareness during fine-tuning, in high-risk fields such as healthcare and law. However, it also requires higher training costs and high-quality labeled data support. For practitioners, the choice should be determined by actual needs: Fine-Tuning emphasizes speed, RAG emphasizes adaptability, while RAFT emphasizes safety and robustness. Whether the application focuses more on real-time interaction, cross-domain transfer, or reliability in high-risk environments will directly determine the choice of method (Fig. 26).

6.2 Architectural trade-offs

6.2.1 Fine-tuning vs. RAG vs. RAFT

Fine-tuning involves the exhaustive retraining of domain-specific datasets, which can be very computationally intensive. This process consists of updating all parameters across the entire model, leading to substantial resource use, especially in larger models. Fine-tuned models will generally demonstrate lower latency during inference as all knowledge is embedded in the model. However, the efficiencies of fine-tuned models about scalability is low

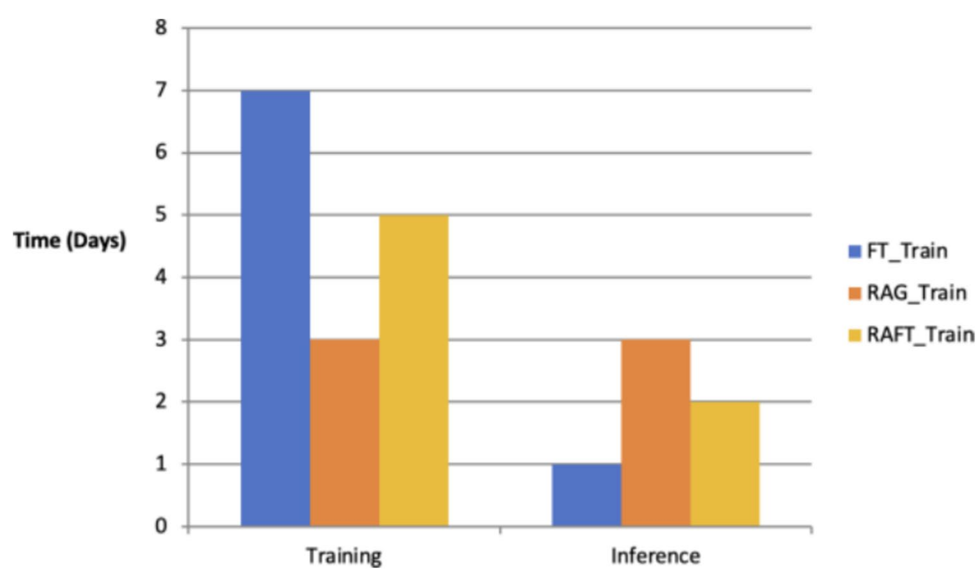

Fig. 26 Training vs. Inference: Fine-tuning, RAG, and RAFT

due to the costs and time involved in retraining when new knowledge is given or when extending the knowledge to additional domains (Ovadia et al. 2024). RAG systems keep the retrieval process separate and independent of the model. The retrieval process can be built and modified independently of the model using an external knowledge base. This modular form factor allows RAG systems

To scale effectively when they are dealing with large or dynamic datasets, as the retrieval process does not go through the same training process as the model and thus can be optimized separately. However, in fluctuating situations, the latency of retrieval may end up lowering system efficiency, and the model's performance will depend on the efficiency of the retrieval process. The issues can be mitigated through optimizations in real-time indexing, caching, and distributed retrieval, but they add layers of complexity to the system (Oracle Cloud Infrastructure 2024). RAFT incorporates retrieval mechanics into the fine-tuning process, so the model can be preconditioned to involve external context during inference. Though RAFT incurs some additional training costs compared to pure RAG, the design of RAFT holds down inference latency as it enables the model to internalize strategies for handling retrieved information. This provides retrieval-aware fine-tuning and allows RAFT to achieve robust performance without requiring constant retraining, which allows it to scale better for specific domains. Additionally, parameter-efficient fine-tuning advances like LoRA make the RAFT process more scalable by minimizing memory and computational requirements, which is beneficial to resource-constrained-environments (Tahir et al. 2024). Figure 27 illustrates the key efficiency factors of RAFT.

6.2.2 Resource efficiency

RAG offers considerable advantages in maintaining up-to-date external information that contributes to factual accuracy in text Generation. However, this further increases inference latency and significantly depends on the performance of the retriever. For example, the generative output can incorporate inaccurate or contradictory information if the retriever returns documents that are either irrelevant to the task or noisy (Abbasiantaeb & Aliannejadi 2024; de Luis Balaguer et al. 2024). In contrast, lavishing the model with domain-specific knowledge through standalone fine-tuning ensures low latency during inference with stable output generation in static contexts. However, this approach can often prove to be inflexible with low-trust characteristics when domain knowledge evolves over time (Ovadia et al. 2024).

Hybrid approaches like RAFT aspire to address trade-offs by incorporating retrieval mechanisms directly within the overall fine-tuning process. This technique augments both internal reasoning and external knowledge utilization by training the models to memorize key domain-specific knowledge and dynamically extract additional context from any retrieved documents. As a result, RAFT can offer robust negative rejection and improved factual accuracy, specifically when relevant external data remains partially noisy (Vidal & Subramanian 2024). Nevertheless, domains with more dynamic and faster knowledge change will usually benefit from retrieval augmented methods that may incur latency costs but provide up-to-date and comprehensive responses.

6.3 Robustness to adversarial inputs

Robustness to adversarial inputs is critical for all three LLM enhancement methods. Traditional fine-tuning encodes domain-specific knowledge into static parameters, making it vulnerable to out-of-distribution adversarial queries, which often cause hallucinations (Lewis et al. 2020). RAG grounds responses with external documents but remains susceptible to retrieval noise and manipulation, limiting its adversarial robustness (Lewis et al. 2020).

RAFT addresses this by integrating retrieval into supervised fine-tuning, exposing models to both relevant and distracting documents to improve context discrimination. Self-RAG (Asai et al. 2024) further enhances robustness through iterative self-reflection, dynamically re-ranking retrieved content to suppress adversarial noise. RAFT thus offers superior robustness compared to fine-tuning or standard RAG approaches.

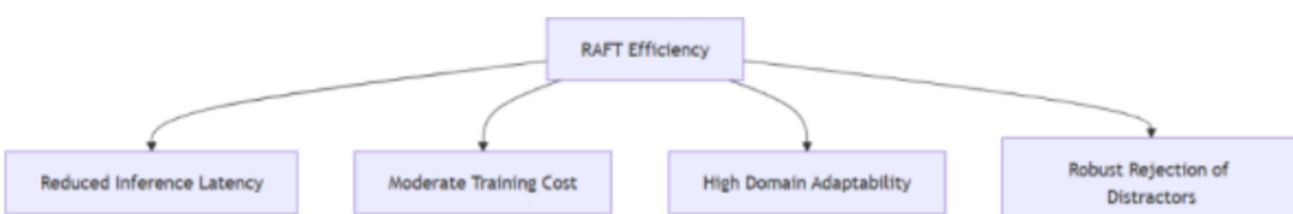

Fig. 27 Key Efficiency Factors of RAFT

7 Challenges and open questions technical challenges

7.1 Data quality and coverage

Although some fine-tuned LLMs have improved their negative rejection ability, there are still barriers that need to be addressed. One major obstacle is the reliance on labeled data for training approaches like NAT and Reinforcement Learning from RLKF, which require large predefined labels to distinguish between safe content and unsafe content. However, labeled data is often scarce or difficult to obtain when it is needed in practical applications, which limits the use of these methods (Li et al. 2020). Another challenge is the method of high demand for computing resources such as NAT and Booster, which is a huge obstacle in real-time or resource-limited applications (J. Liu et al. 2025a, b). This will limit the availability of the method, especially in edge computing scenarios. Researchers are exploring alternative methods to solve these problems, such as self-supervised learning (Chuang et al. 2020) and semi-supervised learning (Shi et al. 2023), which enhance negative rejection ability without relying on extensive annotations. In addition, it is important to improve the adaptability of language models in dynamic environments to ensure that harmful and unsafe content is always rejected.Yet current fine-tuning and RAG pipelines do not directly embed retrieval confidence into the optimisation target; future work should explore a *Retrieval-Aware* DPO framework that treats “insufficient evidence → refuse” as an explicit loss constraint (Lee et al. 2024)

7.2 Computational overhead

However, producing context during the fine-tuning process takes a lot of computational resources. The necessary resources to fine-tune alone are substantial, and adding the retrieval, ranking, and evaluation from an external knowledge source only increases the memory load and time to produce results (Karpukhin et al. 2020). This extra computational burden hinders scalability, specifically in a resource-constrained environment or an environment that requires a real-time output.

7.3 Evaluation metrics and benchmark gaps

Even with much advancement in RAG and fine-tuning methods, current evaluation benchmarks are inadequate to evaluate a model’s capacity to reject adversarial or ambiguous data. Datasets like Natural Questions (Kwiatkowski et al. 2019) and HotpotQA (Yang et al. 2018), allow us to evaluate answer correctness or retrieval recall on a clean dataset but do not measure negative rejection where models refuse to answer due to insufficient evidence or contradictory information. Ribeiro et al. highlighted models may appear accurate overall but still generate misleading outputs from adversarial data (Ribeiro et al. 2020). This gap in evaluation makes it difficult to compare approaches based on their resilience in real-world applications in high-stakes situations.

7.4 Ethical considerations

7.4.1 Bias in retrieval databases

The use of external data raises potential ethical issues, including propagation of bias inherent in the training corpus. If those external docs are misaligned or have an imbalance, then the model might end up rehearsing harmful stereotypes or prompt biased outputs. Furthermore, getting external information involves security and privacy risks, especially when the information is sensitive or proprietary. To address these ethical challenges, robust bias detection and mitigation strategies will have to be used alongside strict data governance protocols to ensure the models are deemed fair and secure.Meanwhile, the PrivLM-Bench study (Li et al. 2024a, b, c) shows that adding differential-privacy filters to the refusal stage can substantially mitigate training-data leakage.

7.4.2 Deployment risks

The wide application of LLMs also brings corresponding security and ethical trends during their deployment process (Bommasani et al. 2021). Prompt Injection attack is a typical threat. Attackers generate responses that deviate from the original meaning through a well-designed input-induced model, thereby avoiding system constraints (Zhang et al. 2024a, 2024b, 2024c, 2024d, 2024e). As pointed out by (Brown et al. 2020), due to the current lack of ability of LLMs to distinguish input sources, Prompt Injection can cause information leakage and even conduct behavioral manipulation without any privilege escalation.

Furthermore, LLMs are also prone to introducing system security vulnerabilities and becoming tools for malicious purposes. For example, attackers can utilize LLMs to generate usable malicious code fragments, thereby assisting in phishing attacks or malware development (Roy et al. 2024). As pointed out by Bai et al. that even with limited input, LLMs can be manipulated to generate highly harmful content (Bai et al. 2022). In this regard, it is necessary to reduce the risks of system deployment through a multi-level security defense system.

7.5 Discussion and implications

This study has contributed to the growing discourse on the security of LLMs by providing a concentrated and systematic analysis of the ability to reject negative inputs. The ability to reject negative inputs is a dimension that has largely been unexplored in previous LLM evaluations, although previous studies have separately examined hallucinations and factual basis (Garcia-Ferrero et al. 2023). Our study views rejection not only as a by-product of improving accuracy, but also as an independent and measurable ability essential for credibility in high-risk applications.

In fact, our comparison results show that RAFT has a significant advantage in rejecting interference and irrelevant inputs. Compared with the fine-tuned baseline, AUPR has increased by 3.22%. This indicates that in future applications, the architecture based on RAFT should be prioritized to reduce the risk of generating erroneous information. More importantly, this work unifies the accuracy of rejection, hallucination mitigation, resource cost, and domain adaptability into an analytical framework. Although earlier work tended to focus on isolated performance metrics, our comprehensive analysis reveals the trade-offs among these dimensions and emphasizes the robustness of RAFT. These insights can not only provide information for model design but also help establish a rejection-aware development pipeline in real-world AI deployments. Finally, Li et al. 's PrivLM-Bench (Li et al. 2024c) demonstrated the necessity of privacy leakage detection in LLM security assessment, laying the foundation for subsequent policy discussions (Li et al. 2024a, b, c). In addition, recent studies indicate that Direct Preference Optimization (DPO) (Wang et al. 2024a, b, c) and real-time attention-flow consistency analysis (Yu et al. 2024) are promising avenues for further improving refusal robustness; these preliminary ideas will be elaborated in the Future Work part of this paper.

8 Future directions

More innovative methods on how LLM frameworks can be made to perform better and also be robust will be explored in future research. In brief, this section discusses several interesting ways to enhance retrieval efficiency; further improve methods and multilingual capabilities for fine-tuning; incorporate external structured knowledge; and construct models that incorporate reasoning paradigms.

8.1 Enhancing retrieval mechanisms

Improvement is possible by enhancing the efficiency and quality of retrieval. Currently, the existing methods are based on intensive retrieval methods, in which the relevant documents are dynamically retrieved. They can, however, lead to delays and computational costs. It was found that advanced dense channel retrieval methods can reduce the cost of retrieval, preserving high accuracy (Karpukhin et al. 2020). Various dynamic caching strategies, vector quantization, and approximate nearest neighbor search are also future work. They will get us the faster and more scalable retrieval, which is very much required for real-time applications. Efficacious retrieval efficiency will not only ensure the speed of reasoning, but also more, it is conducive to the improvement of the performance of the model by means of reliable document selection retrieval.

8.2 Preference-based rejection optimization (DPO Family)

8.2.1 Motivation

Recent work on Direct Preference Optimization (DPO) and its on-policy variants offers a lightweight alternative to full RLHF while preserving alignment quality (Lee et al. 2024; Wang et al. 2024a, b, c). We propose treating refusal as an explicit preference: a correct answer is preferred to a safe refusal, which in turn is preferred to an unsupported or hallucinated answer.

8.2.2 Research Agenda

Preference collection – Curate triple sets ⟨correct answer, justified refusal, unsupported answer⟩ spanning high-stakes domains, where justified refusals are preferred over unsupported answers. Future work will also explore the generation of high-quality synthetic data for refusal examples to augment or replace human-labeled data, addressing data scarcity and enhancing diversity.

Retrieval signal integration – Extend the DPO loss so that "insufficient evidence → refuse" is encouraged whenever retrieved context falls below a confidence threshold; explore joint optimisation with retrieval quality signals.

Evaluation – Report both hallucination rate and over-refusal rate, enabling a balanced view of safety versus helpfulness on established safety benchmarks and the proposed IFC-Bench (see below).

8.3 Information-flow consistency for hallucination control

Layer-wise attention analyses show that, just before a model hallucinates, the influence of retrieved tokens on the generated answer drops abruptly. Yu et al. term this abrupt decline an "evidence gap" (Yu et al. 2024). We take this pattern as evidence that Information-Flow Consistency (IFC)—the degree to which supporting tokens continue to influence later decoding steps—can serve as a practical, model-internal signal for early hallucination detection.

8.3.1 Planned methodology

Diagnostic signal. For every prompt–response pair we will aggregate cross-layer attention weights into a single IFC score. When the score falls below a learned threshold, the sample is flagged as exhibiting an evidence gap.

Mitigation stage. During refusal-aware fine-tuning we will attach either (a) a refusal target or (b) a consistency-regularisation loss to any training instance that triggers the evidence-gap flag. In effect, the model is pushed to refuse—or to realign its attention—whenever its own IFC indicator predicts a hallucination.

Joint optimisation with DPO. Finally, we will integrate the IFC-based penalty into the preference-based Direct Preference Optimization (DPO) objective described earlier. The combined loss encourages the model to produce an answer only when it can maintain high evidence-flow; otherwise it should default to a safe, well-formed refusal.

8.3.2 Why it matters

Because IFC is both differentiable and interpretable, it can be added as a lightweight regulariser that suppresses high-confidence hallucinations without degrading fluency. Moreover, the IFC score itself provides a transparent audit trail, enabling domain experts and regulators to trace whether each answer is genuinely supported by the retrieved evidence or was safely rejected.

8.4 IFC-bench: Bridging the hallucination–over-refusal gap

To track progress on both hallucination and unnecessary refusal, we will release IFC-Bench, a lightweight complement to SafetyBench and OR-Bench.

(1) Task Coverage – Medical consultation, financial compliance, and academic search inquiries.
(2) Label Scheme – Supported answer, justified refusal, unsupported answer, over-refusal.
(3) Artifacts – Public annotation guidelines, evaluation scripts, and layer-wise IFC reference values to ensure reproducibility.

This resource will allow future studies to report hallucination detection, refusal accuracy, and information-flow metrics side-by-side.

8.5 Hybrid approaches (e.g., RAG + reinforcement learning)

Recent studies have explored a hybrid strategy that combines RAG with reinforcement learning (RL) to enhance the robustness and reliability of LLMs. Although RAG improves by providing external evidence during the generation process, it is still vulnerable to the spread of retrieval noise and irrelevant content. To solve this problem, RLHF or synthetic feedback is incorporated into the model based on reward signals for fine-tuning. Applying the RL objective to the selection of retrieval documents can significantly reduce the rate of hallucinations and enhance the consistency of answers (Shinn et al. 2023). Similarly, as proposed by Ram et al., RLAIF (Reinforcement Learning with AI Feedback) to align retrieval enhancement models with human rejection behavior, demonstrating significant benefits in both factual correctness and out-of-scope rejection capabilities (Ram et al. 2023). These hybrid methods take advantage of the benefits of retrieval and adaptive learning, providing a reliable improvement path for LLM systems.

8.6 Cross-domain and multilingual adaptation

Another important research area is refinement and final fine-tuning strategies. LoRA has shown that LLMs can be fine-tuned at a cheaper computational cost without compromising performance (Hu et al. 2022). Work has been done to extend these techniques to support multilingual applications by a few researchers. It can set up strong baselines on cross-language tasks using pre-trained multilingual models, including XLM-R (Conneau et al. 2020) and mT5 (Xue et al. 2021). The developed strong multilingual pre-training, which when combined with RAFT, can generate good models in a cross-language environment ensuring negative rejection capability as well as credibility in low-resource environment. The added multilingual feature will also increase the application of RAFT in such heterogeneous environments.

Limitations and directions. Despite these prospects, evidence availability and alignment remain primary obstacles in low-resource languages: MIRACL and Mr.TyDi document sizable retrieval-effectiveness gaps vs. English, which directly bound RAFT's gains unless the retriever is strengthened and corpora are denoised (Zhang et al. 2021, 2023). We recommend three pragmatic steps: (1) retriever-first—adopt multilingual dense retrievers (e.g., mContriever) and mine bilingual positives and hard distractors from TyDi/MIRACL-style corpora (Izacard et al. 2021; Clark et al. 2020; Zhang et al. 2023); (2) alignment-aware RAFT—encourage explicit grounding to retrieved spans in cross-lingual settings to avoid mislabeling relevant foreign-language passages as distractors (Asai et al. 2021); (3) evaluation—report per-language utility–safety curves (answers vs. refusals) on TyDi/

XOR-TyDi to make the trade-offs explicit, alongside latency deltas from multilingual indexing/translation (Clark et al. 2020; Asai et al. 2021).

8.7 Integration with reasoning and knowledge graphs

RAFT currently relies on unstructured text but could enhance performance by integrating structured external knowledge. Knowledge graphs provide rich semantic relations and multihop reasoning capabilities, offering factual grounding from structured data (W. Liu et al. 2020; Sun et al. 2019). Future work should explore linking RAFT's continuous representations to human-readable knowledge graph structures. For security, GradSafe detects jailbreak prompts and enforces real-time rejection guardrails (Xie et al. 2024). Structured knowledge integration could filter irrelevant information, reduce hallucinations, and improve negative rejection robustness.

9 Conclusion summary of key findings

This review examined the negative rejection ability of LLMs through fine-tuning, RAG, and RAFT. Fine-tuning boosts task performance but struggles to reject irrelevant inputs due to reliance on static knowledge. RAG improves factual grounding with external data but remains sensitive to retrieval noise. RAFT, combining retrieval and fine-tuning, offers stronger rejection robustness and adaptability. Persistent challenges include high computational costs, data quality issues, and the lack of standardized evaluation metrics. Strengthening negative rejection remains essential for safer, more trustworthy AI in practice.

Author contributions All authors contributed to the research concept, design, and methodology. They collectively developed the four-dimensional framework for evaluating negative rejection ability in LLMs and conducted a comparative analysis of fine-tuning, RAG, and RAFT approaches. The systematic literature review, data extraction, and synthesis were carried out by the team. The initial draft of the manuscript was written, and all authors provided critical feedback and revisions on the manuscript's various sections. All authors have read and approved the final manuscript.

Data availability This review article does not involve original data collection or analysis. Therefore, there are no datasets to share. The findings and discussions presented are based on publicly available research studies and datasets referenced throughout the manuscript. For detailed information on these datasets, please refer to the original publications cited in the manuscript.

References

Abbasiantaeb Z, Aliannejadi M (2024) Generate then retrieve: Rethinking conversational response retrieval with llms as answer and query generators. arXiv preprint arXiv:2403.19302. https://doi.org/10.48550/arXiv.2403.19302

Ali W, Zhou X, Shao J (2025) Privacy-preserved and responsible recommenders: from conventional defense to federated learning and blockchain. ACM Comput Surv 57(5):1–35. https://doi.org/10.1145/3708982

Asai A, Kasai J, Clark JH, Lee K, Choi E, Hajishirzi H (2021) XOR QA: Cross-lingual open-retrieval question answering. Proceedings of NAACL-HLT. 489–503. https://doi.org/10.18653/v1/2021.naacl-main.46

Asai A, Wu Z, Wang Y, Sil A, Hajishirzi H (2024) Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024). Retrieved from https://openreview.net/forum?id=hSyW5go0v8

Badampudi D, Wohlin C, Petersen K (2015) Experiences from using snowballing and database searches in systematic literature studies. In Proceedings of the 19th International Conference on Evaluation and Assessment in Software Engineering (EASE 2015), Article No. 17. Association for Computing Machinery. https://doi.org/10.1145/2745802.2745818

Bai Y, Kadavath S, Kundu S, Askell A, Kernion J, Jones A, Chen A, Goldie A, Mirhoseini A, McKinnon C, Chen C, Olsson C, Olah C, Hernandez D, Drain D, Ganguli D, Li D, Tran-Johnson E, Perez E, … Kaplan J (2022) Constitutional AI: Harmlessness from AI feedback (arXiv Preprint No. 2212.08073). arXiv. https://doi.org/10.48550/arXiv.2212.08073

Bengio Y, Mindermann S, Privitera D (2025) International AI Safety Report 2025: Advanced AI Risks and Mitigation, AI Safety Institute. United Kingdom. Retrieved from https://coilink.org/20.500.12592/30e27x2 on 04 Sep 2025. COI: 20.500.12592/30e27x2.

Bommasani R, Hudson DA, Adeli E, Altman R, Arora S, von Arx S, Bernstein MS, Bohg J, Bosselut A, Brunskill E, Brynjolfsson E, Buch S, Card D, Castellon R, Chatterji N, Chen A, Creel K, Davis JQ, Demszky D, … Liang P (2021) On the opportunities and risks of foundation models (arXiv Preprint No. 2108.07258). arXiv. https://doi.org/10.48550/arXiv.2108.07258

Borgeaud S, Mensch A, Hoffmann J, Cai T, Rutherford E, Millican K, van den Driessche GB, Lespiau J, Damoc B, Clark A, de Las Casas D, Guy A, Menick J, Ring R, Hennigan T, Huang S, Maggiore L, Jones C, Cassirer A, … Sifre L (2022) Improving language models by retrieving from trillions of tokens. In K. Chaudhuri, S. Jegelka, L. Song, C. Szepesvári, G. Niu, & S. Sabato (Eds.), Proceedings of the 39th International Conference on Machine Learning (PMLR Vol. 162, pp. 2206–2240). Proceedings of Machine Learning Research. https://proceedings.mlr.press/v162/borgeaud22a.html

Boulesnane A, Souilah A (2024) An evolutionary large language model for hallucination mitigation. In Proceedings of the 1st International Conference on Electrical, Computer, Telecommunication and Energy Technologies (ECTE-Tech 2024) (pp. 1–8). IEEE. https://doi.org/10.1109/ECTE-Tech62477.2024.10851107

Brown TB, Mann B, Ryder N, Subbiah M, Kaplan JD, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, Agarwal S, Herbert-Voss A, Krueger G, Henighan T, Child R, Ramesh A, Ziegler DM, Wu J, Winter C, … Amodei D (2020) Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, & H. Lin (Eds.), Advances in Neural Information Processing Systems, 33 (pp. 1877–1901). Curran Associates. https://arxiv.org/abs/2005.14165

Cao L (2024) Learn to refuse: Making large language models more controllable and reliable through knowledge scope limitation and refusal mechanism. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (pp. 3628–3646). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.212

Chang Y, Wang X, Wang J, Wu Y, Yang L, Zhu K, Chen H, Yi X, Wang C, Wang Y, Ye W, Zhang Y, Chang Y, Yu PS, Yang Q, Xie X, … Yu PS (2024) A survey on evaluation of large language models. ACM Transactions on Intelligent Systems and Technology, 15(3), Article 39, 1–45. https://doi.org/10.1145/3641289

Chaudhary S, Dinesha U, Kalathil D, Shakkottai S (2025) Risk-averse fine-tuning of large language models. Adv Neural Inf Process Syst 37:107003–107038

Chen J, Lin H, Han X, Sun L, AAAI Press (2024) Benchmarking large language models in retrieval-augmented generation. Proceedings of the AAAI Conference on Artificial Intelligence 38:29728. https://doi.org/10.1609/aaai.v38i16.29728

Chen J, Shi W, Fu Z, Cheng S, Li L, Xiao Y (2023) Say what you mean! Large language models speak too positively about negative commonsense knowledge. In: Proceedings of the 61st annual meeting of the association for computational linguistics (Volume 1: Long Papers). Association for Computational Linguistics, pp 9890–9908. https://doi.org/10.18653/v1/2023.acl-long.550

Choi C, Lee Y, Chen A, Zhou A, Raghunathan A, Finn C (2024) AutoFT: Learning an objective for robust fine-tuning (arXiv Preprint No. 2401.10220). arXiv. https://doi.org/10.48550/arXiv.2401.10220

Chuang C-Y, Robinson J, Lin Y-C, Torralba A, Jegelka S (2020) Debiased contrastive learning. In: Larochelle H, Ranzato M, Hadsell R, Balcan MF, Lin H (eds) Advances in Neural Information Processing Systems, 33. Curran Associates, pp 8765–8775

Clark JH, Choi E, Collins M, Garrette D, … Lee K (2020) TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. Transactions of the ACL, 8, 454–470. https://doi.org/10.1162/tacl_a_00317

Conneau A, Khandelwal K, Goyal N, Chaudhary V, Wenzek G, Guzmán F, … Stoyanov V (2020) Unsupervised cross-lingual representation learning at scale. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 8440–8451). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.747

Cui J, Chiang W-L, Stoica I, Hsieh C-J (2025) OR-Bench: an over-refusal benchmark for large language models. In: Proceedings of ICML 2025. https://arxiv.org/abs/2405.20947

Dettmers T, Pagnoni A, Holtzman A, Zettlemoyer L (2023) QLoRA: efficient finetuning of quantized LLMs. In: Advances in neural information processing systems, vol 36. Curran Associates. https://arxiv.org/abs/2305.14314

Dhuliawala S, Komeili M, Xu J, Raileanu R, Li X, Çelikyilmaz A, Weston J (2024) Chain-of-verification reduces hallucination in large language models. In Findings of the Association for Computational Linguistics: ACL 2024 (pp. 3563–3578). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-acl.212

Edge D, Lee Y, Chen A, Zhou A, Raghunathan A, Finn C (2024) From local to global: A graph RAG approach to query-focused summarization (arXiv Preprint No. 2404.16130). arXiv. https://doi.org/10.48550/arXiv.2404.16130

Fang F, Bai Y, Ni S, Yang M, Chen X, Xu R (2024a) En- hancing noise robustness of retrieval-augmented language models with adaptive adversarial training. In Proc. 62nd annu. meeting of the association for computational linguistics (acl 2024). Bangkok, Thailand. https://doi.org/10.18653/v1/2024.acl-long.540

Fang F, Bai Y, Ni S, Yang M, Chen X, Xu R (2024b) Enhancing noise robustness of retrieval-augmented language models with adaptive adversarial training. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024, Volume 1: Long Papers) (pp. 10028–10039). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.540

Farquhar S, Kossen J, Kuhn L, Gal Y (2024) Detecting hallucinations in large language models using semantic entropy. Nature 630(8017):625–630. https://doi.org/10.1038/s41586-024-07421-0

Feldman P, Foulds JR, Pan S (2024) RAGged edges: The double-edged sword of retrieval-augmented chatbots (arXiv Preprint No. 2403.01193). arXiv. https://doi.org/10.48550/arXiv.2403.01193

Gao Y, Xiong Y, Gao X, Jia K, Pan J, Bi Y, Dai Y, Sun J, Wang M, Wang H (2023) Retrieval-augmented generation for large language models: A survey (arXiv Preprint No. 2312.10997). arXiv. https://doi.org/10.48550/arXiv.2312.10997

García-Ferrero I, Altuna B, Álvez J, Gonzalez-Dios I, Rigau G (2023) This is not a dataset: A large negation benchmark to challenge large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 8596–8615). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.531

Gekhman Z, Yona G, Aharoni R, Eyal M, Feder A, Reichart R, Herzig J (2024) Does fine-tuning large language models on new knowledge encourage hallucinations? (arXiv Preprint No. 2405.05904). arXiv. https://doi.org/10.48550/arXiv.2405.05904

Greyling C (2024) RAFT: Combining RAG with fine-tuning. SuperAnnotate Blog. https://www.superannotate.com/blog/raft-retrieval-augmented-fine-tuning

Gupta S et al (2025) Selective self-to-supervised fine-tuning for generalization in large language models (arXiv Preprint No. 2502.08130). arXiv. https://doi.org/10.48550/arXiv.2502.08130

Han Z, Gao C, Liu J, Zhang J, Zhang SQ (2024) Parameter-efficient fine-tuning for large models: A comprehensive survey (arXiv Preprint No. 2403.14608). arXiv. https://doi.org/10.48550/arXiv.2403.14608

Hu, EJ, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, Wang L, Chen W (2022) LoRA: Low-rank adaptation of large language models. In Proceedings of the 10th International Conference on Learning Representations. OpenReview. https://openreview.net/forum?id=nZeVKeeFYf9

Hu Z, Wang L, Lan Y, Xu W, Lim E-P, Bing L, Xu X, Poria S, Lee R (2023) LLM-Adapters: An adapter family for parameter-efficient fine-tuning of large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 5254–5276). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.319

Huang T, Hu S, Ilhan F, Tekin SF, Liu L (2024) Booster: Tackling harmful fine-tuning for large language models via attenuating harmful perturbation (arXiv Preprint No. 2409.01586). arXiv. https://doi.org/10.48550/arXiv.2409.01586

Huber-Flifle, N, Zhang J, Gronvall P, Wei F, Spinelli P (2024) Experimental study of in-context learning for text classification and its application to legal document review in construction delay disputes. In Proceedings of the 2024 IEEE International Conference on Big Data (BigData) (pp. 2119–2125). IEEE. https://doi.org/10.1109/BigData62323.2024.10826061

Izacard G, Grave E, Joulin A (2021) Unsupervised dense information retrieval with contrastive learning (Contriever). arXiv:2112.09118. https://arxiv.org/abs/2112.09118

Izacard G, Lewis P, Lomeli M, Hosseini L, Petroni F, Schick T, … Grave E (2023) Atlas: Few-shot learning with retrieval-augmented language models. Journal of Machine Learning Research, 24(251), 11912–11954.

Jacovi A, Wang A, Alberti C, Tao C, Lipovetz J, Olszewska K, ... Das D (2025). The FACTS Grounding Leaderboard: Benchmarking LLMs' Ability to Ground Responses to Long-Form Input. arXiv preprint arXiv:2501.03200.

Jayawardena L, Yapa P (2024) Improving quality and domain-relevancy of paraphrase generation with graph-based retrieval-augmented generation. In Proceedings of the 10th International Conference on Computing and Artificial Intelligence (ICCAI '24) (pp. 196–208). Association for Computing Machinery. https://doi.org/10.1145/3669754.3669784

Joren H, Zhang J, Ferng C-S, Juan D-C, Taly A, Rashtchian C (2024) Sufficient Context: A New Lens on Retrieval Augmented Generation Systems. The Thirteenth International Conference on Learning Representations. https://openreview.net/forum?id=Jjr2Odj8DJ

Karpukhin V, Oguz B, Min S, Lewis P, Wu L, Edunov S, Chen D, Yih W-T (2020) Dense passage retrieval for open-domain question answering. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (pp. 6769–6781). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.emnlp-main.550

Kitchenham B (2004) Procedures for performing systematic reviews (Technical Report TR/SE-0401). Keele University. https://www.inf.ufsc.br/%7ealdo.vw/kitchenham.pdf

Kwiatkowski T, Palomaki J, Redfield O, … Das D (2019) Natural Questions: A benchmark for question answering research. Transactions of the Association for Computational Linguistics, 7, 452–466. https://doi.org/10.1162/tacl_a_00276

Lee S, Kim S, Yousefpour A, Seo M, Yoo KM, Yu Y (2024) Aligning large language models by on-policy self-judgment. arXiv:2402.11253. https://arxiv.org/abs/2402.11253

Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N et al (2020) Retrieval-augmented generation for knowledge-intensive nlp tasks. Adv Neural Inf Process Syst 33:9459–9474

Li X, Jia X, Jing X-Y (2020) Negative-aware training: be aware of negative samples. In: Proceedings of the 24th European conference on artificial intelligence. IOS Press, pp 1269–1275. https://doi.org/10.3233/FAIA200228

Li H, Guo D, Li D et al (2024a) PrivLM-Bench: a multi-level privacy evaluation benchmark for language models. In: Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long Papers), Bangkok, Thailand. Association for Computational Linguistics, pp 54–73. https://doi.org/10.18653/v1/2024.acl-long.4

Li J, Huang M, Zeng Z, Yu C (2024b) Enhancements in data querying: Applying MMR-integrated in-context learning to LLM-based text-to-SQL. In Proceedings of the 2024 China Automation Congress (CAC) (pp. 4830–4833). IEEE. https://doi.org/10.1109/CAC63892.2024.10865575

Li H, Guo D, Li D, Fan W, Hu Q, Chan C, Yao D, Yao Y, Song Y (2024c) PrivLM-Bench: a multi-level privacy evaluation benchmark for language models. In: Proceedings of the 62nd annual meeting of the association for computational linguistics (Volume 1: Long Papers), Bangkok, Thailand. Association for Computational Linguistics, pp 54–73. https://doi.org/10.18653/v1/2024.acl-long.4

Lin XV, Chen X, Chen M, Shi W, Lomeli M, James R, … Yih W-T (2023) RA-DIT: Retrieval-augmented dual instruction tuning (arXiv Preprint No. 2310.01352). arXiv. https://doi.org/10.48550/arXiv.2310.01352

Liu W, Zhou P, Zhao Z, … Xu X (2020) K-BERT: Enabling language representation with knowledge graph. Proceedings of the AAAI Conference on Artificial Intelligence, 34(3), 2901–2908..

Liu S-Y, Wang C-Y, Yin H, Molchanov P, Wang Y-CF, Cheng K-T, Chen M-H (2024) DoRA: Weight-decomposed low-rank adaptation (arXiv Preprint No. 2402.09353). arXiv. https://doi.org/10.48550/arXiv.2402.09353

Liu J, Qiu Z, Li Z, Dai Q, Zhu J, Hu M, Yang M, King I (2025a) A survey of personalized large language models: Progress and future directions (arXiv Preprint No. 2502.11528). arXiv. https://doi.org/10.48550/arXiv.2502.11528

Liu S, Ye H, Zou J (2025b) Reducing Hallucinations in Large Vision-Language Models via Latent Space Steering. The Thirteenth International Conference on Learning Representations. https://openreview.net/forum?id=LBl7Hez0fF

De Luis Balaguer MA, Benara V, de Freitas Cunha RL, et al (2024) RAG vs fine-tuning: Pipelines, trade-offs, and a case study on agriculture (arXiv Preprint No. 2401.08406). arXiv. https://doi.org/10.48550/arXiv.2401.08406

Luo Q, Ma X, Wei X (2024) Enhancing code generation for dataflow programming: Fine-tuning large language models with the DFCPP dataset. In Proceedings of the 2024 IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA) (pp. 2246–2247). IEEE. https://doi.org/10.1109/ISPA63168.2024.00314

Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE (2025) Hallucination-free? Assessing the reliability of leading AI legal research tools. J Empir Leg Stud 22(2):216–242. https://doi.org/10.1111/jels.12413

Mhammad AF, Agarwal R, Columbo T, Vigorito J (2023) Generative & responsible AI – LLMs use in differential governance. In Proceedings of the 2023 International Conference on Computational Science and Computational Intelligence (pp. 291–295). IEEE. https://doi.org/10.1109/CSCI62032.2023.00051

Min B, Ross H, Sulem E, Pouran Ben Veyseh A, Nguyen TH, Sainz O, Roth D (2023) Recent advances in natural language processing via large pre-trained language models: a survey. ACM Comput Surv 56(2):1–40. https://doi.org/10.1145/3605943

Oliveira VD, Bezerra YF, Weigang L, Brom PC, Celestino VRR (2024) SLIM-RAFT: A novel fine-tuning approach to improve cross-linguistic performance for Mercosur common nomenclature (arXiv Preprint No. 2408.03936). arXiv. https://doi.org/10.48550/arXiv.2408.03936

Oracle Cloud Infrastructure (2024) Scenario 2: Retrieval-Augmented Generation (RAG) benchmarks in generative AI. Oracle Help Center. https://docs.oracle.com/en-us/iaas/Content/generative-ai/scenario-2.htm

Ovadia O, Brief M, Mishaeli M, Elisha O (2024) Fine-tuning or retrieval? Comparing knowledge injection in LLMs. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (pp. 237–250). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.15

Parthasarathy VB, Zafar A, Khan A, Shahid A (2024) The ultimate guide to fine-tuning LLMs from basics to breakthroughs: An exhaustive review of technologies, research, best practices, applied research challenges and opportunities (Version 1.0) [arXiv preprint arXiv:2408.13296]. https://doi.org/10.48550/arXiv.2408.13296

Ram O, Levine Y, Dalmedigos I, Muhlgay D, Shashua A, Leyton-Brown K, Shoham Y (2023) In-context retrieval-augmented language models. Trans Assoc Comput Linguist 11:1316–1331. https://doi.org/10.1162/tacl_a_00605

Rao R (2024). RAG vs fine-tuning: Differences, benefits, and use cases explained. Wevolver. https://www.wevolver.com/article/rag-vs-fine-tuning-differences-benefits-and-use-cases-explained

Ribeiro MT, Wu T, Guestrin C, Singh S (2020) Beyond accuracy: Behavioural testing of NLP models with CheckList. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 4902–4912). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.442

Roy SS, Thota P, Naragam KV, Nilizadeh S (2024) From chatbots to phishbots? Phishing scam generation in commercial large language models. In: 2024 IEEE Symposium on Security and Privacy (SP), San Francisco, pp 36–54. https://doi.org/10.1109/SP54263.2024.00182

Sager N, Cabaza T, Cusack M, Bass R, Dominguez J (2024) Rethinking retrieval-augmented fine-tuning in an evolving LLM landscape. SMU Data Science Review, 8(2), Article 2. https://scholar.smu.edu/datasciencereview/vol8/iss2/2

Salıcı M, Ölçer E (2024) Impact of transformer-based models in NLP: An in-depth study on BERT and GPT. In Proceedings of the 2024 8th International Artificial Intelligence and Data Processing Symposium (IDAP) (pp. 1–6). IEEE. https://doi.org/10.1109/IDAP64064.2024.10710796

Shi Z, Tonolini F, Aletras N, Yilmaz E, Kazai G, Jiao Y (2023) Rethinking semi-supervised learning with language models. In: Findings of the association for computational linguistics: ACL 2023, Toronto, Canada. Association for Computational Linguistics, pp 5614–5634. https://doi.org/10.18653/v1/2023.findings-acl.347

Shi T, Chen K, Zhao J (2024) Safer-Instruct: Aligning language models with automated preference data. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (pp. 7636–7651). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.naacl-long.422

Shinn N, Cassano F, Berman E, Gopinath A, Narasimhan K, Yao S (2023) Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023). Curran Associates. (Original pre-print: arXiv:2303.11366).

Shuster K, Poff S, Chen M, Kiela D, Weston J (2021) Retrieval augmentation reduces hallucination in conversation. In Findings of the Association for Computational Linguistics: EMNLP 2021 (pp. 3784–3803). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.findings-emnlp.320

Soudani H, Kanoulas E, Hasibi F (2024) Fine tuning vs. retrieval augmented generation for less popular knowledge. In Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia-Pacific Region (SIGIR-AP '24) (pp. 12–22). ACM. https://doi.org/10.1145/3673791.3698415

Sulimov D (2024). Prompt-efficient fine-tuning for GPT-like deep models to reduce hallucination and improve reproducibility in scientific text generation using stochastic optimisation techniques (arXiv Preprint No. 2411.06445). arXiv. https://doi.org/10.48550/arXiv.2411.06445

Sun Y, Wang S, Li Y, Feng S, Chen X, Zhang H, … Wu H (2019) ERNIE: Enhanced representation through knowledge integration (arXiv Preprint No. 1904.09223). arXiv. https://doi.org/10.48550/arXiv.1904.09223

Sun X, Xie J, Chen Z, Liu Q, Wu S, Chen Y, Song B, Wang W, Wang L (2025) Divide-Then-Align: honest alignment based on the knowledge boundary of RAG. In: Proceedings of ACL 2025. https://doi.org/10.18653/v1/2025.acl-long.561

Tahir A, Cheng L, Liu H (2024) JORA: JAX tensor-parallel LoRA library for Retrieval-Augmented Fine-Tuning (arXiv Preprint No. 2403.11366). arXiv. https://doi.org/10.48550/arXiv.2403.11366

Thomas DR, Gatz E, Gupta S, Lin J, Tipper C, Koedinger KR (2024) Using generative AI to provide feedback to adult tutors in training and assess real-life performance. In D. Guralnick, M. E. Auer, & A. Poce (Eds.), Creative Approaches to Technology-Enhanced Learning for the Workplace and Higher Education (pp. 204–214). Springer. https://doi.org/10.1007/978-3-031-73427-4_21

Truong TH, Baldwin T, Verspoor K, Cohn T (2023) Language models are not naysayers: An analysis of language models on negation benchmarks. In Proceedings of the 12th Joint Conference on Lexical and Computational Semantics (★SEM 2023) (pp. 101–114). ACL. https://doi.org/10.18653/v1/2023.starsem-1.10

Tural B, Örpek Z, Destan Z (2024) Retrieval-augmented generation (RAG) and LLM integration. In 2024 8th International Symposium on Innovative Approaches in Smart Technologies (ISAS) (pp. 1–5). IEEE. https://doi.org/10.1109/ISAS64331.2024.10845308

Vidal C, Subramanian S (2024) RAFT: A new way to teach LLMs to be better at RAG. Microsoft Tech Community – AI Platform Blog. https://techcommunity.microsoft.com/blog/aiplatformblog/raft-a-new-way-to-teach-llms-to-be-better-at-rag/4084674

Wang R, Li H, Han X, Zhang Y, Baldwin T (2024a) Learning from failure: Integrating negative examples when fine-tuning large language models as agents (arXiv Preprint No. 2402.11651). arXiv. https://doi.org/10.48550/arXiv.2402.11651

Wang Z, Teo SX, Ouyang J et al (2024b) M-RAG: reinforcing large language model performance through retrieval-augmented generation with multiple partitions. In: Proceedings of ACL 2024, pp 2314–2330. https://doi.org/10.48550/arXiv.2405.16420

Wang C, Jiang Y, Yang C, Liu H, Chen Y (2024c) Beyond reverse KL: Generalizing direct preference optimization with diverse divergence constraints. In Proceedings of the Twelfth International Conference on Learning Representations. Retrieved from https://openreview.net/forum?id=2cRzmWXK9N

Warnakulasuriya S, Hapuarachchi K (2024) From knowledge to action: Leveraging Retrieval-Augmented Fine-Tuning (RAFT) to empower quick and confident first-aid decisions in emergencies (Pre-print on ResearchGate). https://doi.org/10.13140/RG.2.2.35911.30888

Wick ML, Liu S, Bartolo M, FitzGerald N, Williams A (2020) Detecting and exorcising statistical demons from language models with anti-models of negative data (arXiv Preprint No. 2010.11855). arXiv. https://doi.org/10.48550/arXiv.2010.11855

Xie Y, Fang M, Pi R, Gong NZ (2024) GradSafe: detecting jailbreak prompts for LLMs via safety-critical gradient analysis. In: Proceedings of ACL 2024, pp 3881–3893. https://doi.org/10.18653/v1/2024.acl-long.30

Xie T, Qi X, Zeng Y, Huang Y et al (2025) SORRY-Bench: systematically evaluating large language model safety refusal. In: Proceedings of ICLR 2025. https://doi.org/10.48550/arXiv.2406.14598

Xu S, Pang L, Yu M et al (2024) Unsupervised information refinement training of large language models for retrieval-augmented generation. https://arxiv.org/abs/2402.18150

Xue L, Constant N, Roberts A, Kale M, Al-Rfou R, Siddhant A, … Raffel C (2021) mT5: a massively multilingual pre-trained text-to-text transformer. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics. ACL, pp 483–498. https://doi.org/10.18653/v1/2021.naacl-main.41

Yang Z, Qi P, Zhang S, Bengio Y, Cohen W, Salakhutdinov R, Manning CD (2018) HotpotQA: A dataset for diverse, explainable multi-hop question answering. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (pp. 2369–2380). ACL. https://arxiv.org/abs/1809.09600

Yu L, Cao M, Cheung JCK, Dong Y (2024) Mechanistic understanding and mitigation of language model non-factual hallucinations. https://arxiv.org/abs/2403.18167

Zhang W, Zhang J (2025) Hallucination mitigation for retrieval-augmented large language models: a review. Mathematics 13(5):856

Zhang X, Ma X, Shi P, Lin J (2023) Miracl: a multilingual retrieval dataset covering 18 languages. Transactions of the ACL 11:1116–1132. https://doi.org/10.1162/tacl_a_00595

Zhang X, Ma X, Shi P, Lin J (2021) Mr.TyDi: A multi-lingual benchmark for dense retrieval. Proceedings of the 1st Workshop on Multilingual Representation Learning (MRL 2021), 127–137. https://aclanthology.org/2021.mrl-1.12

Zhang T, Norouzian A, Mohan A, Ducatelle F (2024a) A new approach for fine-tuning sentence transformers for intent classification and out-of-scope detection tasks. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track (pp. 910–919). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-industry.68

Zhang T, Patil SG, Jain N, Shen S, Zaharia M, Stoica I, Gonzalez JE (2024b) RAFT: adapting language model to domain-specific RAG. In: Proceedings of the First Conference on Language Modeling (COLM 2024). (Paper ID 117). https://doi.org/10.48550/arXiv.2403.10131

Zhang X, Peng B, Tian Y, Zhou J, Jin L, Song L, Mi H, Meng HM (2024c) Self-alignment for factuality: mitigating hallucinations in LLMs via self-evaluation. In: Proceedings of ACL 2024, pp 2759–2774. https://doi.org/10.18653/v1/2024.acl-long.107

Zhang Y, Li Y, Liu J (2024d) Unified efficient fine-tuning techniques for open-source large language models [ResearchSquare pre-print Version 1]. https://doi.org/10.21203/rs.3.rs-4660140/v1

Zhang Z, Lei L, Wu L, Sun R, Huang Y, Long C, Liu X, Lei X, Tang J, Huang M (2024e) SafetyBench: evaluating the safety of large language models. In: Proceedings of ACL 2024. Association for Computational Linguistics, pp 15537–15553. https://doi.org/10.18653/v1/2024.acl-long.830

Zhu R, Jiang X, Wu J, Ma Z, Song J, Bai F, Lin D, Wu L, He C (2025) GRAIT: gradient-driven refusal-aware instruction tuning for effective hallucination mitigation. In: Findings of the association for computational linguistics: NAACL 2025, pp 4006–4021. https://doi.org/10.18653/v1/2025.findings-naacl.223

Ziaei R, Schmidgall S (2023) Language models are susceptible to incorrect patient self-diagnosis in medical applications [arXiv pre-print arXiv:2309.09362]. https://doi.org/10.48550/arXiv.230